Borrowing Constraints and the Rise of the Private Rental Sector*

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Abstract

Borrowing constraints may prevent individual households from buying a home, but do they also reduce the homeownership rate and expand the private rental sector? We develop an empirical strategy that accounts for equilibrium spillovers, based on an assignment model in which investors convert owner-occupied homes into rentals for credit-constrained households. This framework identifies both constrained households and those affected by investment spillovers, allowing quasi-experimental evidence to be interpreted as aggregate effects. Applying the method to Dutch mortgage policy tightenings from 2012 to 2016, we find that borrowing constraints significantly expanded the private rental sector, accounting for up to 75% of its growth through 2017. Our findings imply that borrowing constraints play a central role in shaping tenure outcomes and housing market composition.

Keywords: borrowing constraints, housing tenure, arbitrage, buy-to-let investment, assignment models.

JEL classifications: R31, R21, G51.

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1 Introduction

In the aftermath of the financial crisis, many advanced economies have implemented borrowing constraints as part of broader macroprudential policy frameworks (Cerutti et al., 2017). While such constraints can enhance macroeconomic stability, they may also impede households' access to homeownership. Borrowing-constrained households are often compelled to remain in the rental sector, thereby forgoing the potential wealth accumulation associated with housing capital gains, self-amortizing mortgages, and government subsidies aimed at promoting homeownership. As a result, homeownership rates may decline, while the share of the private rental sector within the total housing stock increases. Indeed, the private rental sector has expanded in many advanced economies in recent years.[1] This trend stands in contrast to the widely held policy objective across advanced countries of encouraging homeownership.[2]

We examine whether the tightening of borrowing constraints—specifically, mortgage payment-to-income (MPTI) constraints—contributes to the expansion of the private rental sector. Recent research has highlighted the central role of MPTI constraints in shaping mortgage debt dynamics and housing prices in the United States (Bhutta and Ringo, 2021; Bosshardt et al., 2024; Defusco et al., 2019), the United Kingdom (Peydró et al., 2023), and Ireland (Acharya et al., 2022; Higgins, 2024). These constraints are also increasingly being adopted in other countries.[3] It is well established that borrowing constraints can lead households whose decisions are restricted to opt for renting over owning (Linneman and Wachter, 1989). However, identifying a causal relationship between borrowing constraints and the size of the private rental sector presents two key empirical challenges.

The first empirical challenge lies in identifying which households are borrowing-constrained. Even when researchers have access to rich financial data at the household level, determining whether a constraint is binding typically requires counterfactual information—that is, knowledge of the housing choices a household would have made in the absence of borrowing constraints.

The second empirical challenge is to identify the aggregate effects of borrowing constraints on the housing market using cross-sectional variation between constrained and unconstrained households. While natural experiments can offer causal insights by exploiting this variation, such analyses are often affected by spillover effects. In particular, borrowing constraints may exert downward pressure on

[1] See Acolin et al. (2016); Gabriel and Rosenthal (2015); Lambie-Hanson et al. (2022) for evidence from the U.S.; Byrne (2020) for the UK, Ireland, and Spain; and Figure 1 for the Netherlands.

[2] See OECD (2023) for an overview of national policies supporting mortgage borrowers. The Netherlands—the focus of this study—allocates more than 1.1 per cent of its GDP to tax relief for homeowners (OECD, 2025).

[3] The Online Appendix of Cerutti et al. (2017) shows that by 2013, 26 of 119 countries had implemented a maximum debt-to-income ratio, up from just 4 countries in 2000.

house prices (Acharya et al., 2022; Akinci and Olmstead-Rumsey, 2018; Duca et al., 2011; Higgins, 2024), which in turn can relax previously binding constraints and enable some households to (re)enter the homeownership market. This feedback mechanism complicates efforts to aggregate individual-level treatment effects: the difference-in-differences estimates between constrained and unconstrained households before and after a policy change may not simply scale up to macro-level outcomes. As a result, quasi-experimental approaches often face a "missing intercept" problem when used to estimate the aggregate impact of borrowing constraints on homeownership rates and the size of the private rental sector.

To address both empirical challenges, we develop a novel assignment model. This model underpins our empirical strategy by identifying households that are *directly* affected by the introduction of mortgage payment-to-income constraints, as well as those *indirectly* affected through equilibrium spillovers. Crucially, it also delineates a control group consisting of households that are unaffected—neither directly nor indirectly—by the policy change. Viewed through the lens of our assignment model, our causal estimates can be interpreted as capturing the aggregate effect of borrowing constraints on the size of the private rental sector.

In our model, households with different income levels compete for a fixed number of houses that vary in quality. The richest households occupy the highest-quality homes, and house prices adjust to allocate lower-quality homes to lower-income buyers. The introduction of MPTI constraints drives a wedge between some households' willingness to pay and their ability to borrow, creating market segments with constrained households and lower prices. This wedge opens up arbitrage opportunities for buy-to-let (or keep-to-let) investors. Because rental payments are not constrained by the ability to borrow, investors can buy houses at depressed prices and rent them to borrowing-constrained households. Investors thus enable constrained households to spend more on housing as tenants than they could as owners.

Investor competition bids up house prices, constraining additional households and drawing in even more investment in the process. This dynamic continues until all arbitrage opportunities are exhausted. We consider two scenarios. In the first, benchmark scenario, households have no preference for owning, and arbitrage opportunities are only exhausted when house prices return to their pre-constraint levels. Because prices eventually coincide with their pre-constraint levels, transactions prior to the introduction of borrowing constraints provide key information on which households become constrained, including those affected through equilibrium spillovers. Our assignment model shows that these spillovers do not bias comparisons between the housing choices of constrained and unconstrained households. Moreover, these pre-policy transactions reveal which homes each household type would select post-reform, even in the absence of detailed household characteristics. In the second scenario, we allow for a positive prefer-

ence for owning. We show that such a preference depresses prices compared to no preference for owning, and limits arbitrage.

We apply the insights from this model to examine the introduction and tightening of MPTI constraints in the Netherlands between 2011 and 2016. Before August 2011, lenders could easily deviate from MPTI guidelines; after this date, the constraints became binding. Between 2012 and 2016, the MPTI constraint was further tightened each year. Figure 1 illustrates this gradual tightening, alongside a corresponding expansion of the private rental sector—patterns that are consistent with the model's predictions.

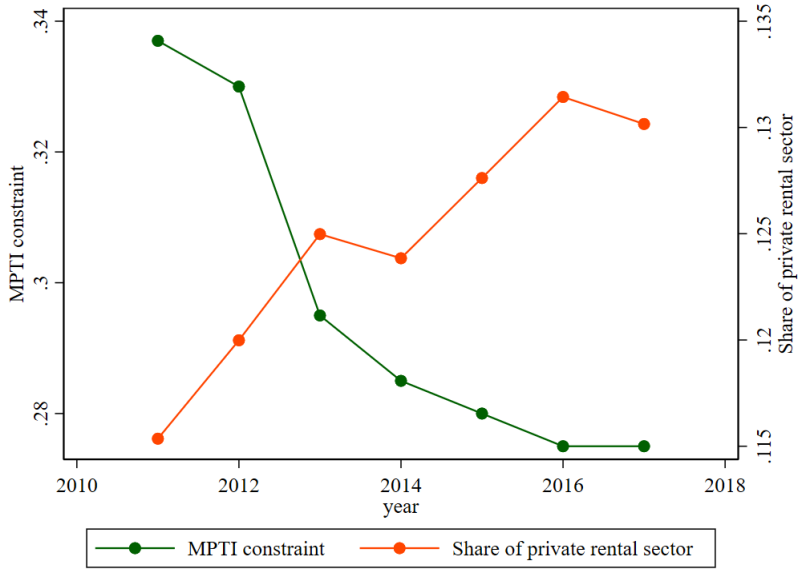


Figure 1: Changes in Mortgage Payment-to-Income Constraints and the Expansion of the Private Rental Sector

Notes: The maximum MPTI varies with household income and mortgage interest rates. The green line shows the MPTI constraint for a household with an annual income of €60,000, evaluated at the average mortgage interest rate in each year. We use interest rates on new mortgage loans with an initial rate-fixation period of 5 to 10 years, which is the most common fixation period in the Netherlands; see De Nederlandsche Bank (DNB). The share of private rental housing is calculated using data from Statistics Netherlands (CBS) StatLine.

The Dutch policy reform offers a particularly suitable setting for studying the effects of borrowing constraints on tenure choice. The reform replaced heterogeneous, lender-specific practices with a uniform MPTI constraint, allowing for clear identification of treated and untreated households. Moreover, the institutional context is advantageous: in the Netherlands, MPTI constraints are the sole relevant borrowing constraint, as neither credit scores nor down-payment requirements exist.[4] Finally, the rich Dutch administrative data covering the full universe

[4]Maximum LTV ratios have always exceeded 100% during our sample period. Although LTV constraints affect leverage (Van Bekkum et al., 2024), we argue in Section 3.1 that these do not affect tenure choice in the Netherlands.

of households, dwellings, and transactions provide us with the required information on household finances and tenure choices.

We examine all transactions by first-time buyers that took place in the two years preceding the enforcement of the mortgage payment-to-income constraint—that is, transactions in 2009 and 2010. For each of these transactions, we assess whether it would have been feasible under the stricter MPTI constraints in place between 2012 and 2017. Consistent with the assignment model, we classify the transacted *houses*, rather than households, into treatment and control groups based on whether and when constraints would have rendered the original transactions infeasible. The intuition is that, assuming the housing stock and income distribution remain relatively stable over time, these houses would continue to be allocated to similar types of households in later years. However, due to the introduction of borrowing constraints, some of these households would no longer be able to purchase the same homes and would be forced to rent instead. The properties that constrained households would otherwise have purchased thus become profitable targets for buy-to-let or keep-to-let investment.

Using a staggered difference-in-differences design, we find that owner-occupied homes are significantly more likely to transition to the rental sector beginning in the exact year that the corresponding 2009 or 2010 transaction would have become infeasible, yet not before. The effects are consistent across both event time and calendar time: an initially owner-occupied home associated with a "constrained transaction" is 0.6 percentage points more likely to be converted to a rental unit.[5] We also find that this increase in rental conversions arises from both buy-to-let and keep-to-let investment, and is concentrated among houses previously purchased by households in the lower part of the income distribution.

Because the effects accumulate over time, the sequence of tightenings—ultimately affecting 17 percent of the 2009–2010 transactions—resulted in a cumulative increase of 1.2 percentage points in the private rental sector by the end of 2017. This represents approximately 15 percent of the total expansion in the sector over the same period. Since the event-study coefficients show no indication of leveling off, we conjecture that additional cumulative effects occurred after 2017 as a result of the 2012–2016 tightenings. However, due to a structural break in the data after 2017, we are unable to estimate these post-period effects.

We proceed by applying three corrections to our empirical estimates. The first two are motivated by extensions to our theoretical framework. In the first extension, we allow households to differ in their preferences for housing relative to other forms of consumption. We show that such preference heterogeneity leads to attenuation bias in our empirical strategy, causing us to underestimate the true effect

[5]Both the two-way fixed effects estimator and the heterogeneity-robust estimator of Callaway and Sant'Anna (2021) yield virtually identical estimates of the average treatment effect on the treated.

of borrowing constraints. Drawing on the approach of Card (1996), we develop a method to correct for this bias and find that the adjusted estimates could be up to 2.2 times larger than the baseline.

The second extension introduces heterogeneity in borrowing constraints arising from differences in household wealth. Some households possess financial assets that can be invested in owner-occupied housing, allowing them to circumvent or partially relax the constraint. This heterogeneity also generates attenuation bias. Using data on household wealth in 2009 and 2010, we refine our predictions of which houses are likely to be converted to rental properties. Incorporating this information increases our estimates by an additional 50 percent.

The third correction addresses selection bias in our sample of transactions. Houses purchased by owner-occupiers in 2009 or 2010 are less likely to reappear on the market in the subsequent years simply because their new owners have recently moved in. Comparing sale probabilities between our sample and the broader set of transacted houses, we find that properties outside our sample are 50 percent more likely to be sold between 2012 and 2017. Accounting for all three corrections, we estimate that borrowing constraints could explain up to 75 percent of the observed increase in the private rental sector.

We conduct two substantive robustness exercises to assess the validity of our empirical findings. First, we address the concern that the sample of houses purchased by first-time buyers in 2009 or 2010 and converted to rental units by 2017 may not be representative of the broader housing stock. To test this, we construct a matched sample using the full registry of residential properties in the Netherlands. For each house purchased by a first-time buyer in 2009 or 2010, we identify a comparable property of the same type (apartment or single-family home), of similar assessed tax value (WOZ value), located on the same street (defined by a 6-digit postcode). We then assign treatment status in this matched sample based on the corresponding treatment assignment in our original sample. Re-estimating our main specifications on this matched dataset, we find that our results remain robust, with somewhat larger estimated effect sizes.

Second, we consider the possibility that MPTI ratios would not have remained stable in the absence of policy-induced borrowing constraints. Following an approach inspired by Bosshardt et al. (2024), we control for a wide range of background trends without taking a stance on the relative importance of any single factor. Specifically, we use a repeated-transaction sample from before the introduction of borrowing constraints to estimate the relationship between buyer income and MPTI ratios. We then use this relationship to predict counterfactual MPTI ratios for 2012–2017, based on the income of new residents of the houses in our baseline sample. These predicted MPTI ratios reflect macroeconomic developments, such as changes in aggregate income, demographic shifts, housing supply, or policies affecting the social housing sector. Any of these changes would alter the income

profile of residents across the housing stock, undermining the assumption that the income of a household in 2009 or 2010 remains a valid proxy for any new resident in subsequent years. We find that our main results are robust to using these predicted MPTI ratios to identify which houses are likely to be converted to rental units.[6]

Finally, we test whether the introduction of MPTI constraints affected house prices. Because individual houses are not transacted every year, we construct a panel of average sales prices for houses of the same type and located on the same street. Consistent with our earlier findings, we observe that a higher fraction of houses assigned to the treatment group—within the same type-street cell—is associated with a greater likelihood of conversion to rental housing. However, the treatment share has no statistically significant effect on average sales prices. This result is consistent with our assignment model and stands in contrast to the findings of Greenwald and Guren (2025) for the United States, who report significant price effects. In the Dutch context, the absence of price effects alleviates concerns that arbitrage in housing markets may be limited (Glaeser and Gyourko, 2007), and supports the modeling assumption that although the housing market consists of a continuum of segments, each segment can flexibly accommodate either owner-occupied or rental housing.

Our paper makes three main contributions. First, we show that buy-to-let investment can arise as an equilibrium outcome driven by borrowing constraints. In contrast to Gete and Reher (2018), our explanation does not depend on downward price rigidity. Instead, it builds on the assignment model of Braid (1981), who analyzes rental markets; in our extension, the user cost of housing replaces rent when the dwelling is owner-occupied. While Landvoigt et al. (2015) examine the effects of a down-payment constraint in a quantitative assignment model, they do not formally analyze how borrowing constraints shape the equilibrium allocation of households across the housing market.[7]

Our second contribution is to demonstrate that the causal effects estimated using a appropriately constructed difference-in-differences design can be interpreted as aggregate effects. In this sense, our hybrid approach to resolving the missing-intercept problem is conceptually similar to the strategy proposed by Wolf (2023) for estimating fiscal multipliers. Our empirical design leverages the assignment model to identify which houses are susceptible to spillovers from borrowing constraints and subsequent buy-to-let investment. Importantly, the model also shows

[6]We do not use observed house prices in 2012–2017 to construct counterfactual MPTI ratios because house prices are endogenous to borrowing constraints. While mortgage interest rates could also influence MPTI ratios and may be considered exogenous in a small open economy such as the Netherlands, the formulation of MPTI constraints in the Dutch context neutralizes this effect; see Section 3.1 for further details.

[7]Other housing assignment models include Epple et al. (2020), Higgins (2023), Määttänen and Terviö (2014), and Määttänen and Terviö (2021).

that such investment does not influence the probability of investment in houses that would be purchased by unconstrained households, even though prices in these segments may rise as well. In the language of the difference-in-differences literature, the stable unit treatment value assumption (SUTVA) holds. As a result, we are able to identify the aggregate effect of borrowing constraints on the size of the private rental sector.[8]

Our third contribution concerns the specific empirical effect of borrowing constraints on the size of the private rental sector in the Netherlands. We find a statistically significant and economically meaningful impact of the introduction and subsequent tightening of the mortgage payment-to-income constraint. Importantly, our estimates are based on a counterfactual derived explicitly from a theoretical assignment model and implemented at the level of individual houses. This contrasts with much of the existing quasi-experimental literature, which typically examines effects at broader geographic levels; see, for example, Carozzi (2019), Gete and Reher (2018), Greenwald and Guren (2025), and Hanson (2023).[9] Complementary to Carozzi (2019), who focuses on keep-to-let investment, and to Mills et al. (2019) and Hanson (2023), who study buy-to-let investment, we find that both channels contribute meaningfully to the expansion of the Dutch private rental sector.

Perhaps most closely related to our work is Thiel and Zaunbrecher (2023), who analyze the gradual relaxation of MPTI constraints for dual earners—but not single earners—in the Netherlands between 2015 and 2019.[10] Like us, they use a difference-in-differences methodology to identify causal effects, but additionally estimate a structural model to evaluate equilibrium outcomes in the rental market.[11]

2 Theory

In this section, we develop a framework to analyze tenure choice in a housing market with borrowing constraints. A central insight is that the effect of these constraints depends critically on whether households prefer homeownership over renting. We therefore consider several cases, gradually increasing in complexity.

We begin by presenting a benchmark assignment model of the housing market in the absence of borrowing constraints. In this frictionless setting, tenure choice

[8] Related structural work that calibrates equilibrium models with MPTI constraints includes Greenwald (2018), Mabille (2022), Ma and Zubairy (2021), and Castellanos et al. (2024).

[9] Another strand of the literature relies on survey data to examine the impact of borrowing constraints; see Duca and Rosenthal (1994) and Fuster and Zafar (2021).

[10] Of course, we also account for the differential treatment of dual and single earners in our empirical analysis.

[11] For work on loan-to-value (LTV) constraints in the Dutch context, see Van Bekkum et al. (2024) and De Veirman and de Jong (2025). Bolliger et al. (2025) and Tzur-Ilan (2023) study LTV constraints and their implications for homeownership in Switzerland and Israel, respectively.

is indeterminate unless households exhibit some preference for owning. To ensure a well-defined equilibrium, we assume that households have at least a slight preference for homeownership.

In subsection 2.3, we introduce borrowing constraints but rule out buy-to-let investment. This case can be interpreted as one where households have a prohibitively strong preference for owning, preventing the emergence of a rental market. In subsection 2.4, we extend the model to allow for buy-to-let investment, enabling the formation of a private rental sector. We first analyze the case in which households are indifferent between owning and renting, and then generalize the model to accommodate intermediate preferences for homeownership.

2.1 Environment

We consider a market populated by a continuum of households with identical preferences over housing services q and other consumption c.[12] Preferences are represented by the utility function

$$u = u(q, c), \tag{1}$$

which is strictly increasing in both arguments, strongly quasi-concave, and twice differentiable. We further assume that the marginal willingness to pay for housing,

$$M(q, c) \equiv \frac{\partial u / \partial q}{\partial u / \partial c}.$$

is increasing in other consumption, i.e.

$$\frac{\partial M(q, c)}{\partial c} > 0. \tag{2}$$

This single-crossing property implies that, for a given house, households with more other consumption are willing to pay more for additional housing services.

To allow for a preference for homeownership, we assume that renting entails a utility cost $\tau \geq 0$, measured in units of other consumption. This disutility of renting may arise from a pure preference for owning, but can also capture other factors—such as differences in tax treatment between owners and renters—in a reduced-form way. Accordingly, renters receive utility $u(q, c - \tau)$, whereas owners receive utility $u(q, c)$.

The housing stock is fixed and consists of a continuum of indivisible houses indexed by their quality $q \in [q^{\min}, q^{\max}]$. The distribution of housing services is given by a strictly increasing and continuously differentiable function $G(q)$, with density $g(q)$. Ownership status (owner-occupied or rental) is determined endogenously in equilibrium for each quality q. The total number of houses is $S = G(q^{\max})$.

Households differ in income y, which represents the resources available for consumption in the period under consideration.[13] The income distribution is given

[12] In Appendix A.5, we extend the model to allow for heterogeneity in preferences.

[13] Appendix A.2 provides a dynamic foundation for this static budget constraint.

by a strictly increasing and continuously differentiable function $F(y)$, with density $f(y)$ and support $[y^{\min}, y^{\max}]$. The total number of households is $B = F(y^{\max})$.

Households maximize utility subject to a budget constraint. For renters, the constraint is

$$c + r(q) = y, \tag{3}$$

where $r(q)$ denotes the rent of a house of quality q. For owners, the constraint is

$$c + p(q) = y, \tag{4}$$

where $p(q)$ denotes the user cost for a house of quality q. The user cost is linked to the transaction price $P(q)$ according to

$$p(q) = \gamma P(q) - E(\Delta P(q)), \tag{5}$$

where γ captures financing costs, maintenance, insurance, and taxes, and $E(\Delta P(q))$ denotes the expected capital gain. In what follows, we take user costs as the relevant equilibrium objects without explicitly modeling their components.[14] User costs $p(q)$ may be nonlinear, so that the marginal price of housing dp/dq may vary with q.[15]

We assume that the number of houses does not exceed the number of households, $B \geq S$. Consequently, not all households can be accommodated in the primary housing market. We therefore introduce an outside rental option, consisting of housing services $q^* \leq q^{\min}$ at rent r^*, which is available to all households.

Households decide whether to participate in the primary housing market by comparing the utility from market housing with the outside option, which yields

$$u^*(y) = u(q^*, y - r^* - \tau). \tag{6}$$

Because households with higher incomes have a higher willingness to pay for housing and q^* is inferior to all houses in the market, it is natural that only sufficiently high-income households participate in the primary housing market; an intuition that is formalized below. The participation threshold y^P is defined implicitly by

$$B - F(y^P) = S, \tag{7}$$

so that exactly S households enter the housing market in equilibrium. In the following subsections, we characterize equilibrium allocations and tenure outcomes under various assumptions on borrowing constraints and investment.

[14] Out-of-pocket housing expenses, such as mortgage payments and maintenance, may exceed user costs when house prices are expected to appreciate. Borrowing constraints typically apply to these monetary expenses rather than to user costs.

[15] In some cases, the function $p(q)$ will not be everywhere differentiable.

2.2 An assignment model without borrowing constraints

In this subsection, we characterize the equilibrium allocation of houses and the associated user cost schedule in the absence of borrowing constraints. We first determine the assignment of households to houses and then derive the user cost function that supports this allocation. When households are indifferent between owning and renting, tenure status is indeterminate. We therefore focus on the case in which households experience a (possibly small) disutility from renting, $\tau > 0$, so that all houses are owner-occupied in equilibrium.

User costs adjust to clear the housing market. Households with income below the participation threshold y^P choose the outside option, while those with income above y^P compete for the available houses. In equilibrium, the marginal participant—i.e., the household with income y^P—must be indifferent between consuming the lowest-quality house $q^{\min}$ and the outside option:

$$u\left(q^{\min}, y^P - p(q^{\min})\right) = u^*(y^P). \tag{8}$$

This condition determines the user cost of the lowest-quality house, $p(q^{\min})$. For any given house, households with higher incomes have more other consumption. By the single-crossing property, they therefore have a higher marginal willingness to pay for housing services. Combined with market clearing, this implies an assortative equilibrium in which higher-quality houses are assigned to higher-income households. Intuitively, any allocation in which a lower-income household occupies a higher-quality house than a higher-income household cannot be sustained in equilibrium, because the two households could profitably exchange houses at an intermediate price.

The following lemma formalizes this assignment. Proofs are provided in Appendix A.1

Lemma 1 (Assignment rule)**.** *In equilibrium, the user cost function $p(q)$ is strictly increasing and continuous in q, and the assignment follows the continuous function*

$$y(q) = F^{-1}\left(F\left(y^P\right) + G(q)\right). \tag{9}$$

The equilibrium assignment therefore preserves the income ranking: households with higher incomes occupy higher-quality houses. Given this assortative assignment, the user cost function must ensure that each household finds its assigned house optimal. This optimality condition implies that the slope of the user cost schedule equals the marginal willingness to pay for housing, and thereby characterizes the equilibrium user cost function, as formalized in the following lemma.

Lemma 2 (Equilibrium user costs)**.** *Let $y(q)$ be given by the assignment rule in (9). In equilibrium, the user cost function $p(q)$ is uniquely characterized by*

$$\frac{dp}{dq} = M(q, y(q) - p(q)) \tag{10}$$

for all $q \in (q^{\min}, q^{\max})$*, together with the boundary condition that* $p(q^{\min})$ *is determined by* (8)*.*

We can thus trace out the equilibrium user cost schedule. Starting from the critical income y^P, the lowest housing quality $q^{\min}$, and the corresponding user cost $p(q^{\min})$, the assignment rule (9) determines the income associated with each housing quality, while (10) determines the corresponding user cost.

2.3 Borrowing constraints

We now introduce mortgage payment-to-income constraints. In particular, we impose that for all households the user cannot exceed a fraction μ of income:[16]

$$p(q) \leq \mu y. \tag{11}$$

To characterize the constrained equilibrium, it is convenient to express user costs as a function of household income rather than housing quality. In the unconstrained equilibrium, the assignment rule (9) defines a continuous and strictly increasing relationship between housing quality and income, allowing us to write its inverse as $q(y)$. This inverse provides a natural benchmark for analyzing the effect of the constraint. The corresponding user cost schedule is $p(q(y))$. Since the equilibrium user cost is increasing in housing quality and housing quality is increasing in income, $p(q(y))$ is increasing in income, with slope

$$\frac{dp}{dy} = \frac{dp}{dq}\frac{dq}{dy} = \frac{dp}{dq}\frac{f(y)}{g(q(y))}. \tag{12}$$

From now on, we denote the unconstrained and constrained user cost schedules by $p^U(q)$ and $p^C(q)$, respectively. Throughout, we express both schedules as functions of income using the unconstrained assignment rule $q(y)$. The shape of the unconstrained user cost schedule $p^U(q(y))$ determines how the mortgage payment-to-income constraint affects the equilibrium. We first consider a simple and empirically relevant case, and consider other cases later.

In this simple case, illustrated in Figure 2, the outside option and critical income y^P are such that the unconstrained user cost of the lowest-quality house exceeds the mortgage payment-to-income constraint, i.e., $p^U(q^{\min}) > \mu y^P$. Moreover, in this case, the unconstrained user cost schedule (shown in black) is concave and eventually increases less rapidly than borrowing capacity (dashed line), such that there exists a unique income level y' at which the unconstrained user cost schedule intersects the mortgage payment-to-income constraint: $p^U(q(y')) = \mu y'$.

A natural conjecture is that, in the constrained equilibrium, the user cost schedule coincides with the borrowing limit for all $y \in [y^P, y']$, and with the unconstrained user cost schedule for $y > y'$. This, however, cannot be an equilibrium. By

[16] In Appendix A.4 we generalize this analysis to an arbitrary distribution of maximum user costs.

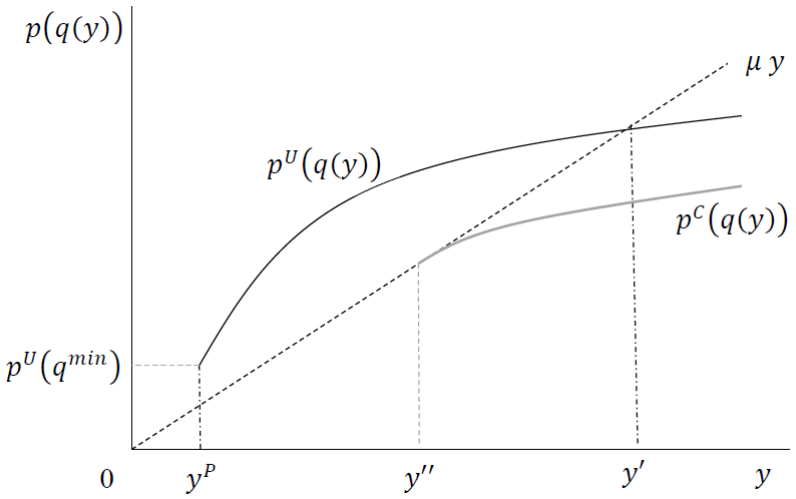


Figure 2: Mortgage Payment-to-Income Constraints and the User Cost of Housing under High $p(q^{\min})$

(12), the slope of the unconstrained user cost schedule at y' satisfies $dp^U/dy < \mu$. Thus, at y', borrowing capacity increases more rapidly with income than households' willingness to pay for housing. The mortgage payment-to-income constraint therefore cannot be binding at that point. Consequently, there must exist an income level $y'' < y'$ at which the borrowing constraint ceases to bind. Since marginal willingness to pay is continuous in income, the constrained user cost schedule must also join the unconstrained schedule smoothly. Rather than exhibiting a kink at y'', it gradually bends away from the borrowing limit, as illustrated by the gray curve in Figure 2.[17]

Figure 3 illustrates a slightly more complex case. In contrast to the previous case, the user cost of the lowest-quality house is now below the mortgage payment-to-income constraint: $p^U(q^{\min}) < \mu y^P$. Hence, the constraint is slack at the bottom of the income distribution. However, the unconstrained user cost schedule subsequently crosses the borrowing limit from below. This gives rise to an additional income threshold, denoted by y^*, at which the mortgage payment-to-income constraint first becomes binding: $p^U(q(y^*)) = \mu y^*$. For incomes below y^*, the borrowing constraint is slack; for incomes just above y^*, it binds.

Having characterized the constrained equilibrium in two illustrative cases, we are now in a position to describe it in full generality. The empirically relevant case is the simple case discussed above; the following characterization is included to make clear how the equilibrium extends when the borrowing constraint becomes binding over multiple income intervals. We first consider the case in which the borrowing constraint is initially slack at the participation threshold y^P, and then turn to the case in which it already binds at that threshold.

[17]The limiting case in which the constrained user cost schedule departs immediately from the borrowing constraint is illustrated in Appendix A.3.

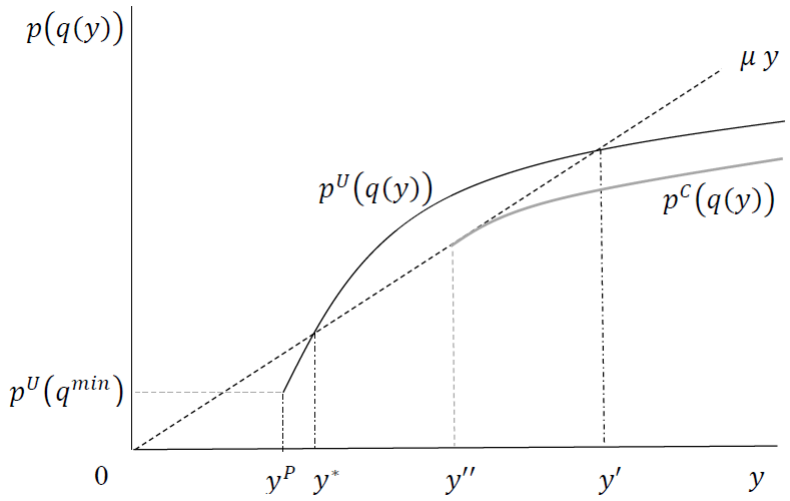


Figure 3: Mortgage Payment-to-Income Constraints and the User Cost of Housing under Low $p(q^{\min})$

Proposition 1. *Suppose that the borrowing constraint is slack at the participation margin, $p^U(q^{\min}) \leq \mu y^P$, and let y^* denote the smallest income $y \geq y^P$ such that the unconstrained user cost schedule violates the borrowing constraint immediately to the right of y, that is,*

$$p^U(q(\tilde{y})) > \mu\tilde{y} \quad \text{for all } \tilde{y} > y \text{ sufficiently close to } y.$$

If no such income exists, the constrained equilibrium coincides with the unconstrained equilibrium: $p^C = p^U$ and the assignment rule remains $q(y)$.

Otherwise, the constrained equilibrium also preserves the assignment rule $q(y)$. Before y^*, the constrained schedule coincides with the unconstrained schedule:

$$p^C(q(y)) = p^U(q(y)), \qquad y \in [y^P, y^*].$$

From y^* onward, the constrained user cost schedule is constructed recursively as follows.

1. *The schedule follows the borrowing constraint as long as, evaluated at $p^C(q(y)) = \mu y$, the marginal-willingness-to-pay slope is at least as large as the slope of the borrowing constraint:*

$$M(q(y), (1-\mu)y)\frac{f(y)}{g(q(y))} \geq \mu.$$

   Let y'' denote the smallest income $y \geq y^*$ at which this inequality is violated, if such an income exists.

2. *If such an income y'' exists, the constrained user cost schedule is characterized by*

$$\frac{dp^C(q(y))}{dy} = M(q(y), y - p^C(q(y)))\frac{f(y)}{g(q(y))}$$

   with initial condition

$$p^C(q(y'')) = \mu y''.$$

3. *If the constrained schedule reaches the borrowing constraint again, steps 1 and 2 are repeated.*

Proposition 1 shows that the constrained user cost schedule is obtained by alternating between two regimes. As long as households would like to spend at least the maximum permitted fraction μ of their income on housing, the borrowing constraint binds and the user cost schedule follows the borrowing constraint. Once households' marginal willingness to pay for housing falls below the marginal increase in borrowing capacity, the borrowing constraint becomes slack and the user cost schedule is again determined by the equilibrium condition equating marginal willingness to pay to the marginal user cost. If the resulting schedule subsequently reaches the borrowing constraint again, the same construction is repeated.

The constrained user cost schedule therefore coincides with the unconstrained schedule up to the first binding point y^* and weakly lies below it thereafter, with strict inequality whenever the borrowing constraint binds over an interval of positive length. Since the assignment rule is unchanged, households with incomes below y^* are unaffected. Households with incomes above y^* pay a lower user cost for the same house and are therefore strictly better off. The intuition is that the borrowing constraint limits the ability of constrained households to bid up the price of housing. This reduction in user costs propagates upward through the equilibrium schedule, benefiting even households for whom the borrowing constraint is eventually slack.

We now turn to the case in which the borrowing constraint already binds at the participation threshold.

Corollary 1. *Suppose that the borrowing constraint already binds at the participation margin,*

$$p^U(q^{\min}) > \mu y^P.$$

Then the constrained equilibrium preserves the assignment rule $q(y)$ and is obtained from the recursive construction in Proposition 1 by replacing y^* with y^P.

The simple case illustrated in Figure 2 is a special case of Corollary 1. The limiting case in which $y'' = y^P$ is illustrated in Appendix A.3. In that case, households' marginal willingness to pay for housing is already below the slope of borrowing capacity at the participation threshold. Consequently, the constrained user cost schedule coincides with the borrowing constraint only at y^P and immediately departs from it thereafter.

The above characterization assumes that all houses remain owner-occupied. In the next section, we investigate whether this equilibrium is robust to the entry of buy-to-let investors.

2.4 Buy-to-let investors

So far, the market consisted exclusively of owner-occupied housing. We now allow for the entry of absentee buy-to-let investors. Whether borrowing constraints create profitable opportunities for these investors depends crucially on households' willingness to substitute renting for owning.

We model buy-to-let investors as competitive absentee landlords with deep pockets who are not subject to borrowing constraints. An investor purchasing a house of quality q earns expected profits

$$\Pi(q) = r(q) + E\Delta P(q) - \gamma P(q). \tag{13}$$

Investors face the same cost of capital and other costs as owner-occupiers, as specified in (5), and enter whenever $\Pi(q) > 0$. Combining (13) with (5) shows that zero profits require

$$r(q) = p(q). \tag{14}$$

Thus, free entry among investors links rents directly to the user costs implied by house prices.

Without borrowing constraints, households are willing to pay at most $p^U(q)$ in rent for a house with user cost $p^U(q)$. Equation (14) then implies zero investor profits, so there is no profitable entry. However, borrowing constraints can create a wedge between these two objects. As shown in the previous section, they depress user costs while constrained households would be willing to pay more for additional housing. Investors can exploit this wedge if households are willing to rent.

The preference for homeownership determines whether they are. If τ is sufficiently large, the maximum rents households are willing to pay do not support profitable buy-to-let investment, and the owner-occupied equilibrium characterized in the previous section remains valid. We first consider the opposite polar case, $\tau = 0$, in which households are indifferent between owning and renting. We return to intermediate values of τ in Section 2.5.

Proposition 2. *Suppose that households are indifferent between owning and renting ($\tau = 0$). Then there exists an equilibrium with buy-to-let investment in which*

1. *the assignment of households to housing qualities remains $q(y)$;*
2. *households for whom $p^U(q(y)) > \mu y$ rent, while households for whom $p^U(q(y)) \leq \mu y$ remain owner-occupiers;*
3. *equilibrium rents in the rental sector satisfy $r(q(y)) = p^U(q(y))$;*
4. *equilibrium user costs in the owner-occupied sector satisfy $p(q(y)) = p^U(q(y))$.*

Proposition 2 follows from an arbitrage argument. In the owner-occupied equilibrium with borrowing constraints, user costs are below $p^U(q)$ wherever borrowing constraints affect prices. When $\tau = 0$, households can express their willingness to pay for additional housing in the rental market. Whenever a household is willing to pay a rent exceeding the prevailing user cost, (14) implies that an investor can earn positive profits at the prevailing house price. Investor entry therefore bids up house prices and, through (5), user costs.

This arbitrage eliminates any wedge between rents and user costs. It stops at the unconstrained schedule: at $p(q) = p^U(q)$, households' maximum willingness to pay in rent is $p^U(q)$, while (14) requires $r(q) = p(q)$. Investor profits are therefore zero. Households for whom $p^U(q(y)) > \mu y$ cannot purchase their assigned house at these user costs and rent it instead; all other households can remain owner-occupiers. Since both groups face the unconstrained housing expenditure schedule, the assignment remains $q(y)$. This establishes Proposition 2.

Figure 2 illustrates the resulting expansion of the set of borrowing-constrained households. Without buy-to-let investment, the borrowing constraint binds only up to y''. Investor entry raises house prices and user costs until they reach the unconstrained schedule, so households with incomes between y'' and y' also become unable to purchase their assigned houses. In equilibrium, all households for whom

$$p^U(q(y)) > \mu y \tag{15}$$

rent. Thus, although borrowing constraints initially depress house prices, investor arbitrage restores the unconstrained user cost schedule and shifts the adjustment entirely to tenure. Because $\tau = 0$, household welfare is also identical to the unconstrained benchmark.

2.5 Intermediate preferences for homeownership

The previous section considered the polar case $\tau = 0$, in which households are indifferent between owning and renting and buy-to-let investors fully arbitrage away the effect of borrowing constraints on user costs. At the other extreme, if τ is sufficiently large, households are unwilling to rent at any rent that makes buy-to-let investment profitable, and the owner-occupied equilibrium with borrowing constraints remains valid. We now consider the case of intermediate preferences for homeownership.

In this intermediate case, a renter is willing to pay less than an owner for the same house. In particular, a household with income y assigned to quality $q(y)$ is indifferent between owning and renting if the rent satisfies

$$r^\tau(q(y)) = p^U(q(y)) - \tau.$$

Thus, buy-to-let investors can charge at most the unconstrained user cost less the compensation required for the disutility of renting.

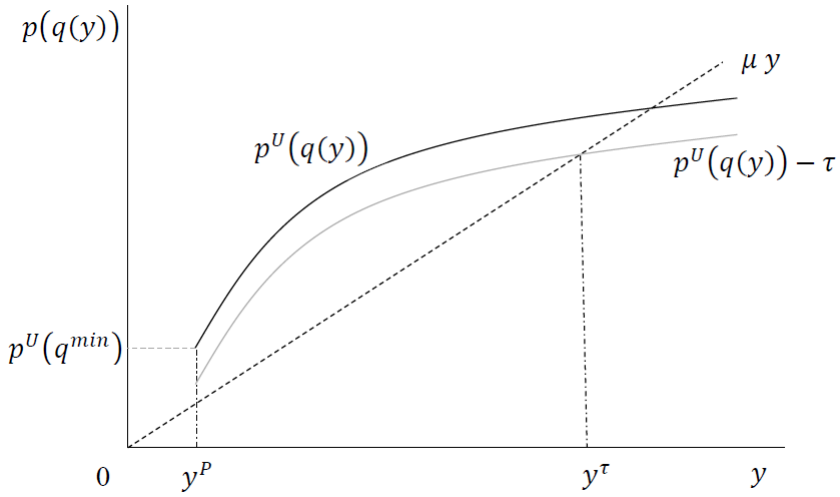


Figure 4: The mortgage payment-to-income constraint and the rent schedule that leaves households indifferent between renting and owning.

We focus on the empirically relevant case in which the borrowing constraint binds at the participation threshold and the rent schedule that leaves households indifferent between renting and owning, $p^U(q(y)) - \tau$, intersects the borrowing constraint exactly once. Let y^τ denote this intersection:

$$p^U(q(y^\tau)) - \tau = \mu y^\tau.$$

Figure 4 illustrates the threshold income y^τ. The equilibrium characterized below builds on this threshold, which separates two regions with different implications of the borrowing constraint.

To avoid the additional notation associated with multiple binding intervals, we proceed as follows. Let $y'' \geq y^\tau$ denote the first income at which, along the borrowing constraint,

$$M(q(y), (1-\mu)y)\frac{f(y)}{g(q(y))} < \mu,$$

if such an income exists. If no such income exists, the user cost schedule remains on the borrowing constraint for all $y \geq y^\tau$. If such an income exists, we impose that the owner-occupied user cost schedule does not return to the borrowing constraint after it has departed from it:

Assumption 1 (No second binding interval)**.** *The solution to*

$$\frac{dp(q(y))}{dy} = M(q(y), y - p(q(y)))\frac{f(y)}{g(q(y))}$$

with initial condition

$$p(q(y'')) = \mu y''$$

does not intersect the borrowing constraint again for $y > y''$.

Proposition 3. *Suppose that $\tau > 0$ is such that y^τ exists as defined above and, if y'' exists, Assumption 1 holds. Then there exists an equilibrium with buy-to-let investment in which*

1. *households with incomes below y^τ rent, whereas households with incomes at and above y^τ remain owner-occupiers;*
2. *the assignment of households to housing qualities remains $q(y)$;*
3. *rents in the rental segment satisfy*

$$r(q(y)) = p^U(q(y)) - \tau, \qquad y \in [y^P, y^\tau);$$

4. *the user cost schedule for owner-occupiers is as follows. If y'' exists,*

$$p(q(y)) = \mu y, \qquad y \in [y^\tau, y''],$$

and for $y > y''$

$$\frac{dp(q(y))}{dy} = M(q(y), y - p(q(y)))\frac{f(y)}{g(q(y))}$$

with initial condition $p(q(y'')) = \mu y''$.

Otherwise,

$$p(q(y)) = \mu y \qquad \text{for all } y \geq y^\tau.$$

The proposition shows how preferences for homeownership limit the arbitrage performed by buy-to-let investors. When $\tau = 0$, investors bid house prices up until user costs coincide with the unconstrained schedule. When $\tau > 0$, households require compensation for renting. This compensation reduces the maximum rent investors can charge and therefore the house prices they are willing to pay. Arbitrage is consequently incomplete: user costs remain below the unconstrained schedule, while rents in the rental sector satisfy

$$r(q(y)) = p^U(q(y)) - \tau < p^U(q(y)).$$

The welfare implications combine those of the previous two sections. Tenants attain the same utility as in the unconstrained owner-occupied equilibrium because the rent exactly compensates them for the disutility of renting. Owner-occupiers, by contrast, continue to benefit from user costs that lie weakly below the unconstrained schedule while consuming the same housing quality.

As τ increases, the rent schedule $p^U(q(y)) - \tau$ shifts downward, so the threshold income y^τ moves upward and the rental sector contracts. In the limit, if τ is sufficiently large, no rent that households are willing to pay supports profitable buy-to-let investment, and the equilibrium converges to the owner-occupied equilibrium with borrowing constraints characterized in Section 2.3. The extent to which house prices remain below the unconstrained benchmark therefore reflects the strength of households' preference for homeownership.

3 Empirical approach

For each level of τ, the assignment model clearly identifies which households are constrained to borrow, and which houses investors will convert to rental houses. In this section, we explain how we can leverage these insights to establish aggregate effects from cross-sectional variation. We exploit two sets of results.

First, Proposition 2 and 3 show that the assignment in the equilibrium with borrowing constraints and buy-to-let investment coincides with the assignment in the equilibrium without borrowing constraints. This result implies that the incomes of the people who live in certain houses before the introduction of borrowing constraints (ex ante) are informative about the incomes of the residents of those houses after the introduction of borrowing constraints (ex post).

Second, Proposition 2 shows that if households have no preference for owning, the user cost (or rent) function in the equilibrium with borrowing constraints and buy-to-let investment coincides with the user cost function in the equilibrium without borrowing constraints. This result implies that the mortgage payments of the people who live in certain houses ex ante are informative about the mortgage payments that the residents of those houses ex post pay or *would* pay if they were owner-occupiers (without a preference for ownership). Moreover, Proposition 3 shows that the rent function falls by exactly τ if households do have a preference for owning.

Our empirical analysis starts with the benchmark in which households have no preference for ownership. Combining the results above, in this case the mortgage payment-to-income (MPTI) ratios ex ante are informative about the MPTI ratios of residents ex post, including those counterfactual ratios that cannot be observed because residents are now renters. As a result, the MPTI ratios of the residents of the houses transacted ex ante can be directly compared to the MPTI constraint. Those ratios that violate the constraint identify the houses that in the absence of the constraint would be bought by first-time buyers, but that in the presence of borrowing constraints will be targeted by investors for conversion to the rental sector.

It may be surprising that despite the presence of equilibrium spillovers, ex-ante MPTI ratios are informative about ex-post counterfactual MPTI ratios.

For our baseline estimates, we proceed as if the assignment of treatment status follows directly from the simple model in the previous section. However, we also consider the empirical implications of two extensions to our model—heterogeneity in borrowing constraints, and heterogeneity in preferences—that we discuss in Appendices A.4 and A.5, respectively. In the presence of such heterogeneity, we may misclassify some treated and control-group houses, resulting in an attenuation bias Aigner (1973). Based on Card (1996), we propose a method to correct for such attenuation bias. In addition, we use information on wealth to arrive at a tighter characterization of who faces binding borrowing constraints, and who does not.

Finally, in a robustness exercise, we account for the incomes of new residents to address the possibility that the assignment between households and houses has changed during our sample period. Such a change may be due to changes in preferences, the housing stock or the income distribution.

We apply our method to study the introduction and tightening of the MPTI constraint in the Netherlands. Studying this policy reform and setting has a number of benefits. First, it allows us to determine which constraints apply to which households. Even without macroprudential policy, mortgage lenders will generally tie the size of mortgages to the ability of borrowers to service their mortgage debt. The policy reform replaced such lender-specific practices with a uniform constraint. Moreover, in the Netherlands, MPTI constraints are the only relevant borrowing constraints. Although potentially access to mortgages is limited by three different types of borrowing constraints—loan-to-value (LTV) or down-payment constraints, debt service-to-income or mortgage payment-to-income constraints, and credit scores (Acolin et al., 2016)—neither credit scores nor down-payment requirements exist in the Netherlands. In particular, maximum LTV ratios have always exceeded 100% during our sample period. Finally, MPTI constraints in the Netherlands are conditional on interest rates. In the next section, we provide the institutional details to this policy reform and discuss other housing policy reforms during our sample period.

3.1 Institutional background

In the year 1989 mortgage lenders in the Netherlands introduced a Code of Conduct. An important element of this Code of Conduct for mortgage loans is that households' maximum mortgage loan is tied to households' income. As a rule of thumb, net mortgage payments should not exceed roughly 30% of net income. However, the exact percentage is determined by an independent institute for expenditure research, the National Institute for Family Finance Information (NIBUD), which annually releases a detailed table indicating the maximum mortgage payment-to-income ratio (MPTI) as a function of the mortgage interest rate and gross income.[18] Because the NIBUD targets net mortgage payments as a share of net income—accounting for the relatively generous mortgage interest deductibility—this 'NIBUD rule' rises as mortgage interest rates decline. During our sample period, falling interest rates contributed to rising house prices. However, in the absence

[18]Before 2012, if a mortgage loan was provided to more than one borrower, the applicable maximum MPTI was based on the gross income of the borrower with the highest income. However, the mortgage lender could take the joint incomes of the borrowers into account when determining their borrowing capacity, i.e., gross household income constituted the denominator of the MPTI. In 2012, the gross income of the secondary earner started to be taken into account also for the determination of the applicable maximum MPTI, but first only for 33%. In 2016 this percentage was raised to 50%, and in 2017 to 60%, see Thiel and Zaunbrecher (2023) for more details. Of course, we always apply the relevant criteria in our empirical analysis.

of additional policy tightenings or changes in net income, maximum net monthly mortgage payments would have remained stable.

During our study period, the NIBUD rule has always been a formal qualification requirement for the national mortgage guarantee (NHG). In addition, the Code of Conduct of Dutch mortgage lenders referred to the NIBUD rule as a guideline. However, before 2011, when consumers did not want to buy NHG insurance, mortgage lenders could simply exceed the NIBUD rule if they informed consumers about the associated risks and documented the justification for the exceedance.[19] As can be seen in Figure 5, before 2011 high mortgage payment-to-income ratios were indeed very common. Common justifications for the exceedance of the NIBUD rule were the expectation for higher-educated households to experience higher-than-average income growth, or plans for home renovation that would increase its value.

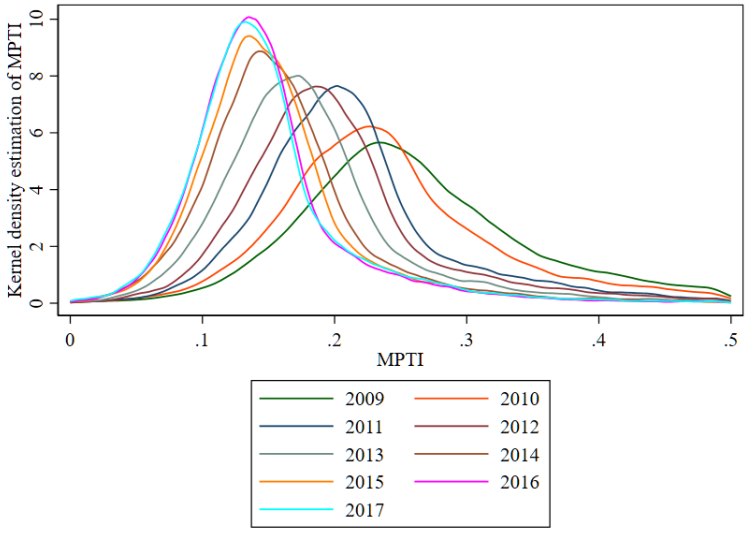


Figure 5: Distribution of Mortgage Payment-to-Income Ratios across Years

Notes: This figure presents kernel density estimates of the mortgage payment-to-income (MPTI) ratio for first-time homebuyers from 2009 to 2017. Observations with an MPTI ratio equal to zero or greater than 0.5 are excluded.

As announced on March 21, the Code of Conduct changed on August 1, 2011. From then onwards, the possible exceptions to the NIBUD rule were listed and only referred to mortgage refinancing, bridge loans, and additional mortgage financing for energy-saving measures or purchasing the most energy-efficient houses (thus excluding general renovation plans). In addition, it was formalized that future income could only be considered when the borrower's employer would provide an unconditional written commitment to that future income, to be effective within six months. Finally, even the determination of income of self-employed

[19] See Appendix D for the relevant sections in the translated policy documents, before and after 2011.

consumers left little wiggle room. In 2013, parts of the Code became elements of national Dutch law.

On top of the exclusion of previously common exceptions, in the years between 2011 and 2016 the NIBUD rule was tightened every consecutive year. Figure 6 shows the evolution of the maximum MPTI for four gross income levels from the start of 2008 until the end of 2021. The figure shows a relatively large decrease in the maximum MPTI in January 2013, and smaller but still significant changes in January of the next three years. In 2017 and later years, there were modest relaxations of the maximum MPTI for some income levels.

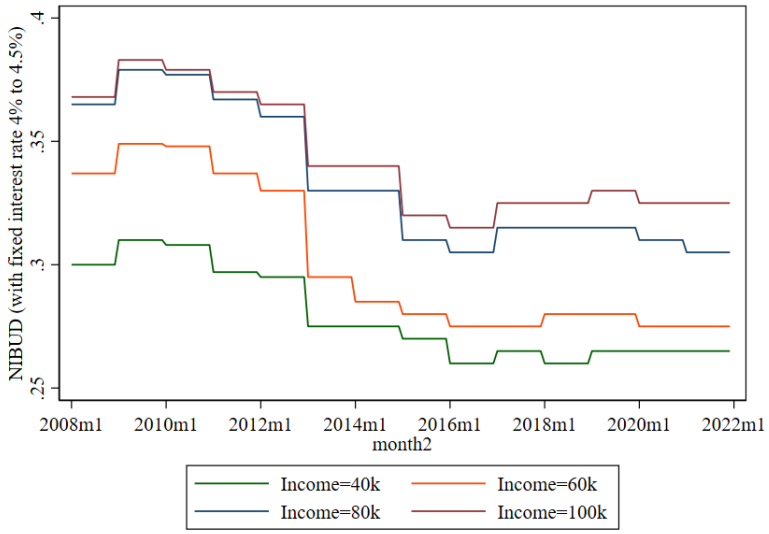


Figure 6: Evolution of the NIBUD MPTI Constraint at a Fixed Interest Rate

Notes: This figure shows the evolution of the maximum mortgage payment-to-income (MPTI) ratio prescribed by NIBUD, holding the mortgage interest rate fixed. We use the 4.5%–5% interest-rate bracket, which corresponds to the prevailing mortgage rates in 2010 and 2011. The data are obtained from NIBUD.

During our study period, three other housing market policy reforms that may affect our results were introduced in the Netherlands. First, on June 15, 2011, the government reduced the transaction tax from 6% to 2%. While this reform could potentially affect house prices and borrowing behavior, its effect was compensated by a second reform: the introduction of a loan-to-value constraint on August 1, 2011, in the same updated Code of Conduct as discussed above. Until 2011, households could realize an initial LTV that was substantially higher than 1, and a year before the reform the median LTV at origination stood at 1.1 (Van Bekkum et al., 2024). Mortgage loans could be used to finance additional costs associated with purchasing a home, and one important item that could be financed with a mortgage was the transaction tax. The Code of Conduct that became effective on August 1, 2011, stipulated a maximum LTV of 1.06. Because this maximum LTV was followed by many homebuyers, the median LTV at origination also fell to 1.06 (Van Bekkum et al., 2024). In other words, the median LTV at origination fell by

exactly the same 4 percentage points as the reduction of the transaction tax.

In addition, between 2013 and 2018, the maximum LTV was lowered further to 100% in annual steps of 1 percentage point. Consequently, until the end of our study period in 2017, the maximum LTV was always higher than 1. However, a mortgage with an LTV higher than 1 does not allow the borrower to exceed the NIBUD rule for a maximum MPTI. The tightening of the maximum LTV thus did not reduce the maximum size of the mortgage loan that a household with a given income could obtain; it only increased the fraction of the loan that should be spent on the house itself, rather than on any additional costs. For that reason, if MPTI constraints would not have been tightened, when a household with a certain income could buy and finance a house with a mortgage loan before the reduction of the maximum LTV, it could still buy and finance this house with a mortgage loan after the reduction of the maximum LTV. And similarly, when it would not be able to buy and finance this house before the tightening of the LTV constraint, it would still not have been able to afterwards. We therefore argue that the tightening of the LTV constraint did not affect tenure choice, even though of course it affects leverage, as shown in Van Bekkum et al. (2024).

The third policy reform was a 2013 change in the eligibility for the mortgage interest deductibility that is relatively generous in the Netherlands. To take maximum advantage of mortgage interest deductibility, mortgage types that postponed repayment of the loan had become very popular in the Netherlands. Investment mortgages that combined an interest-only mortgage with a life insurance that paid out at the end of the term of the mortgage (commonly referred to as savings mortgages) were popular since the 1980s, but in the 1990s interest-only mortgages (without an accompanying investment obligation) became the most popular type. However, from 2013 onwards, new mortgages were only eligible for mortgage interest deductibility if these were repaid as a 30-year annuity mortgage or faster. As a result, the savings and interest-only mortgages lost their appeal and annuity mortgages became the norm again.[20]

However, even before the reform, the maximum mortgage loan prescribed by the NIBUD rules was based on a 30-year annuity mortgage, meaning that the maximum mortgage loan that people could take out was not affected by the reform. Households opting for an interest-only mortgage before 2013 realized lower mortgage payment-to-income ratios, but the maximum size of their loan was still determined as if they had chosen a 30-year annuity mortgage. In fact, if it were not for the accompanying tightening of MPTI constraints, MPTI ratios should rise as the result of the deductibility reform, because after the reform households would have to pay for amortization. However, as can be seen in Figure 5, MPTI ratios continued to decline in 2013. We conclude that only the tightening of MPTI ratios

[20] Bernstein and Koudijs (2024) show that the induced amortization led to additional savings for affected households.

is relevant for tenure choice.

The combination of uniform enforcement and lower maximum MPTI ratios imply that before August 2011 housing transactions were realized that would have been impossible afterwards. Based on the assignment model, we derive the hypothesis that the houses involved in such transactions had a larger probability of being converted to the rental sector than other houses. We test this hypothesis using a difference-in-difference design.

3.2 Data

We rely on the micro-data from Statistics Netherlands to construct our sample. Our baseline sample consists of all initially owner-occupied houses bought by first-time buyers in the Netherlands in the years 2009 and 2010. The years 2009 and 2010 were two years in which the NIBUD-rule was relatively stable, and these were the last two years before the change in the Code of Conduct was announced. We ensure that the first-time buyers buy these houses for self-occupation by retaining only those houses that are still owner-occupied by the end of 2010.[21] We present descriptive statistics of our sample of houses in Table A1 in Appendix B. In the same appendix, we also present more details about our data sources and how sample selection changes the number of observations.

Next, we compute the MPTI ratios of the buyers of these houses. We observe their mortgage loan (ML) at the end of the year of purchase, and given the popularity of interest-only mortgages and the small amortization at the initial stage of an annuity mortgage, we assume it equals the mortgage loan at origination. We also assume that the mortgage maturity is 30 years, or 360 months, as is the norm in the Netherlands. Then we define the monthly mortgage payment (MP) according to

$$MP = ML * \frac{r(1+r)^{360}}{(1+r)^{360}-1}, \tag{16}$$

in which r denotes the monthly interest rate.[22] Finally, we compute MPTI ratios by dividing the monthly mortgage payment by monthly gross household income. The MPTI distributions shown in Figure 5 for all years between 2009 and 2017 have been computed analogously.

Figure 7 presents the mortgage payments as a function of the gross household income of our sample of first-time buyers in 2009 and 2010. The figure also shows the maximum mortgage payments as prescribed by the NIBUD rules for the years

[21] Due to data limitations we can only ensure that buyers are first-time buyers to the extent that they were no owners between 2003 and 2009/2010.

[22] Following practice of the Dutch banks, we compute monthly interest rates by dividing annual interest rates by 12. We do not have access to individual interest rates, but we assume that individual interest rates are equal to the average interest rate in the month of purchase, using interest rates on new mortgage loans with an initial rate fixation period of 5 to 10 years, as are most common in the Netherlands, see De Nederlandsche Bank (DNB).

from 2011 to 2017. As such, this figure is the empirical analogue of Figure 2. By comparing the mortgage payments in 2009 and 2010 to the NIBUD rule in subsequent years, we can determine which transactions would no longer have been possible due to borrowing constraints. We remove those transactions that violate the NIBUD rule already in 2011 (the 'always treated'), leaving us with a sample of almost 50k unique houses.

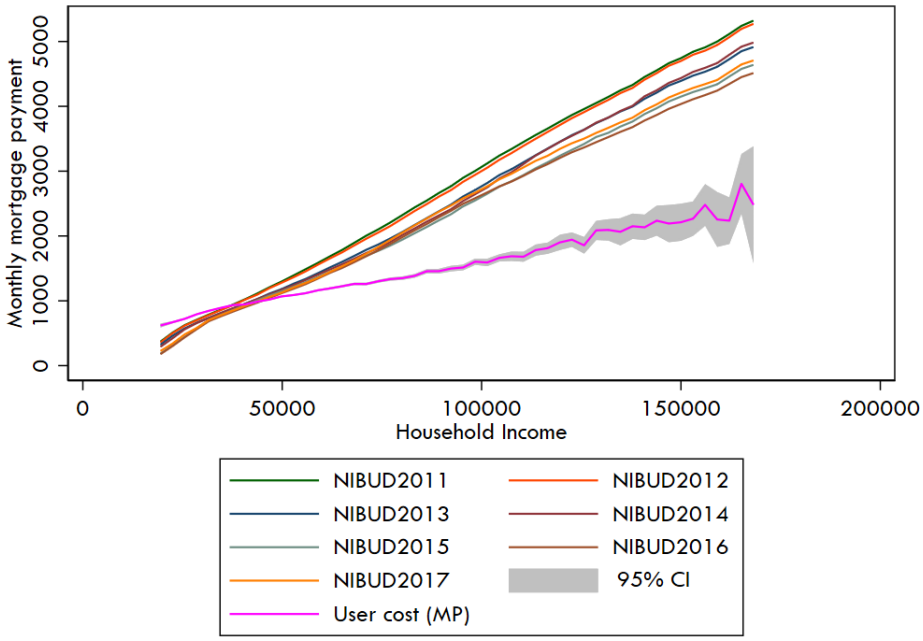


Figure 7: Maximum and Observed Mortgage Payments by Household Income

Notes: This figure compares the maximum monthly mortgage payments implied by the NIBUD rules for 2011–2017 with observed monthly mortgage payments for housing transactions in 2009 and 2010, plotted against gross household income.

We transform our sample of houses into a panel dataset by adding for each subsequent year whether the original 2009 or 2010 transaction would have been possible, or would violate the new borrowing constraints. We complete the panel by adding information on the ownership status of the houses in our sample in the period from 2011 to 2017. We choose this time frame because the data on ownership status suffer from a structural break due to a revision in its statistical methods in 2011 and again in 2018, making data before 2011 and after 2017 non-comparable.

Figure 8 shows the share of the transactions observed in 2009 and 2010 that violate the NIBUD rules in later years. It shows a continuing deepening and staggered timing of treatment, with the exception of 2017. As can be seen in Figure 6

In 2017, a relaxation of the NIBUD rule led to some observations exiting the treatment group (see Figures 1 and 8). Such treatment reversals could introduce estimation bias. To mitigate this bias, we exclude those observations that reversed treatment in 2017.

The increase is particularly large in 2013, which is in line with the decrease of

the NIBUD rule. As NIBUD gradually declined and more households were constrained, the average MPTI of sold homes also gradually declined between 2011 and 2017. The kernel density distribution of MPTI across years in Figure 5 confirms the pattern. From 2009 to 2017, the distribution of MPTI gradually shifted leftward. Transactions with relatively high MPTI became unfeasible due to NIBUD regulations.

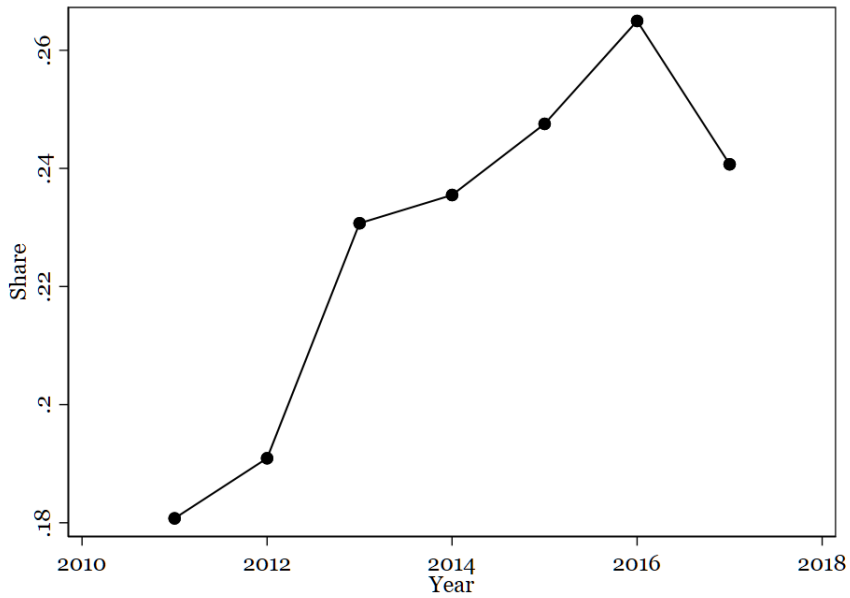


Figure 8: Share of Properties Violating the NIBUD Constraint over Time

Notes: This figure shows the annual share of properties classified as violating the NIBUD mortgage payment-to-income constraint. It illustrates both the evolution of the treatment group over time and the timing of treatment exposure across years.

3.3 A staggered difference-in-differences approach

For those transactions for which this rule was satisfied at the time of realization, we could moreover check if they would still have been in accordance with the NIBUD rule in the years that followed (until 2017) if everything else would have remained constant. That is, we ask the question if the same transaction would have been feasible if the income of the purchasing household, the price of the property and the mortgage interest rate would still be the same as in 2009 or 2010, while we use the NIBUD rules of later years. This provides us with a useful granular indicator of a binding borrowing constraint for households that are the likely buyers of such houses, which does not depend on market developments after 2011. According to our theory exactly these houses are exposed to buy-to-let activity.

We use a panel for the years 2011-2017 in our baseline analyses, which ownership status is observed for every house in every year. Ownership status of houses is indicated in our data as a categorical variable with categories: (i) owned by a

housing association, (ii) owner-occupied, (iii) owned by a private landlord, and (iv) unknown. Houses in the third category can be owned by a private person as well as by a firm, for instance, an institutional investor. We define a dummy variable to indicate whether the house is a private rent house in that year. Since our sample was all owner-occupied in late 2010, we actually saw a shift from owner-occupied to private rent. In the period considered here, there are many switches between private rental and owner-occupied and *vice versa*. As noted above, many of the switches from owner-occupied to private rental resulted from buy-to-let or keep-to-let activities, while many switches in the other direction were from less expensive rent-controlled housing to owner-occupation.

We define whether a house enters the treatment group in a year based on whether the house's fixed MPTI (in 2009 or 2010) violates the NIBUD rule of that year. What we are concerned about is whether buyers with the same income and the same interest rate can still buy a house of the same price when the NIBUD rule changes. When controlling for year fixed effects, it can also be interpreted as, we assume that there is a common trend in the growth of interest rates, housing prices, income, etc. over years. All houses were originally in the control group (in 2010) and were owner-occupied. With the gradual decline of the NIBUD rule, many houses began to enter the treatment group, and some houses were converted to private rentals. In other words, we define the year in which a house first violates the NIBUD rule as the treatment year.

Our identification relies on the assumption that the MPTI of the same house across years remains constant, as bought by the household with the same income. To check the MPTI stability assumption, we compare the annual average difference and standard error of house MPTI from 2006 to 2017 with the MPTI of the same house when it was re-traded in 2010. Figure 9 shows that from 2006 to 2010, the MPTI of the same house remained relatively stable, with cross-year differences being small, at most around 0.02. However, after 2011, due to the gradually tightening credit constraints, the MPTI of the same house decreased year by year. This implies that using the 2009 and 2010 MPTI as the counterfactual for 2011–2017 is reliable, despite the presence of measurement error. In the following sections, we consider a series of correction methods to address potential biases in the MPTI stability assumption.

We document heterogeneous effects of borrowing constraints across income groups. Figure 10 describes the distribution of household income of the entire sample and treated sample. This suggests that households subject to borrowing constraints are concentrated among low-income households, which has a perfect match with our assignment model. Correspondingly, growth in the private rental sector between 2011 and 2017 was also concentrated among low-income groups. Figure 11 documents that the probability of entering the private rental sector increases by 4 to 6 percentage points for low-income people below the 20th per-

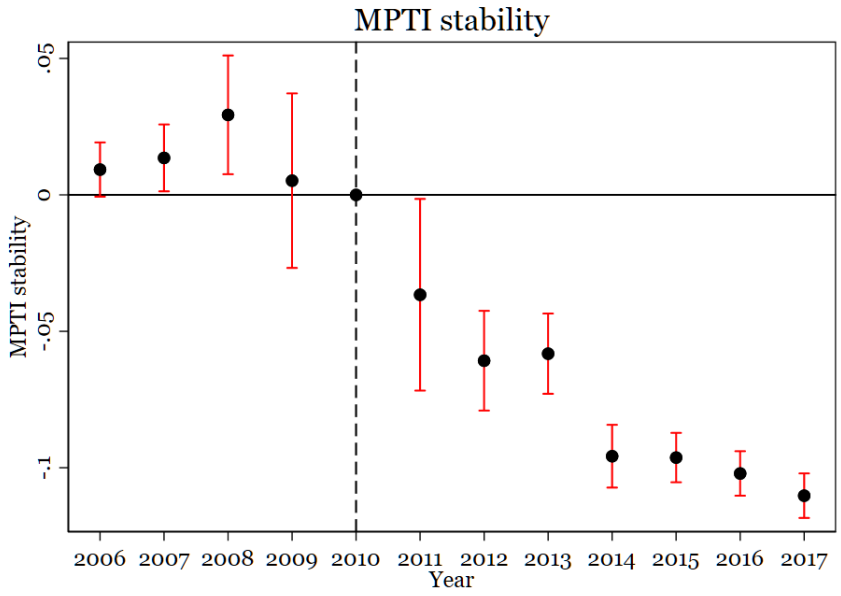


Figure 9: Within-Property Differences in MPTI Relative to 2010

Notes: This figure examines the stability of MPTI for the same property over time. We first identify properties transacted in 2010 that also had a repeat transaction in a given year between 2006 and 2017. For each year, we pair those repeat transactions with the corresponding property's 2010 transaction and calculate the difference in MPTI between the two transactions. Each coefficient therefore represents the average within-property difference in MPTI relative to 2010. The vertical bars indicate 95% confidence intervals.

centile. The probability of entering the private rental sector decreases with increasing income and disappears at the 50th percentile. These figures provide suggestive evidence that the growth of the private rental sector is likely due to borrowing constraints. Figure A6 illustrates how the growth in the private rental market share diverges between the treatment and control groups in our sample. The treatment group consists of properties that violated the maximum MPTI limit between 2011 and 2017, while the rest form the control group. At the end of 2010, all properties were owner-occupied. Over time, the treatment group exhibited a faster change in the share of private rental housing. By the end of 2017, the share of private rental housing in the treatment group was 8.86%, compared to 6.82% in the control group.

4 Empirical results

4.1 Baseline estimation

We employ the DID strategy at the property-year level. With the panel dataset constructed in the previous section, we start our estimation with a canonical two-

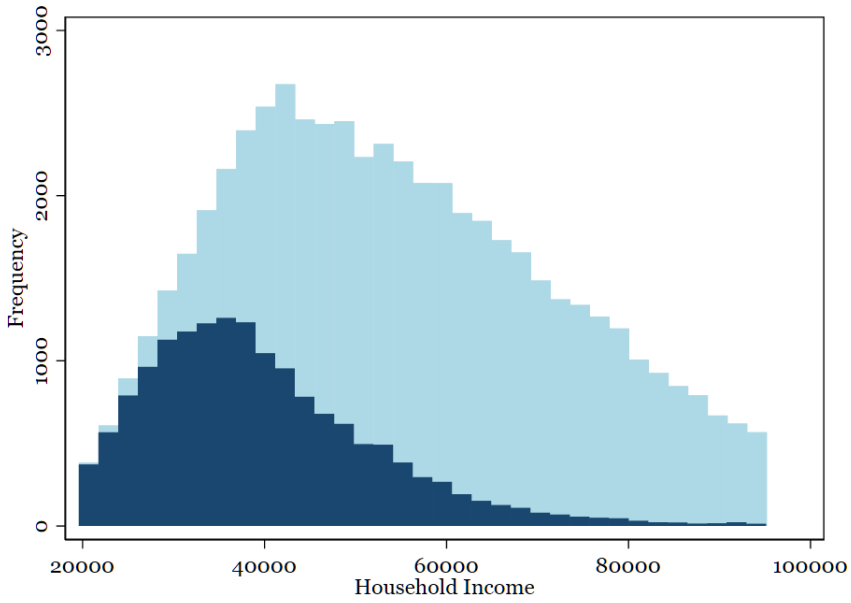


Figure 10: Household Income Distributions in the Baseline and Constrained Samples

Notes: This figure compares the distribution of household income in the baseline sample with that of households subject to credit constraints between 2011 and 2017. The lighter bars represent the baseline sample, while the darker bars represent the constrained sample.

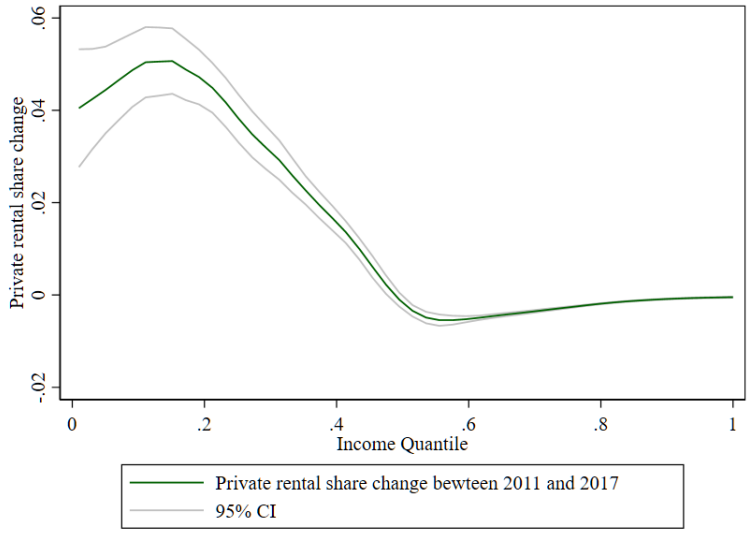


Figure 11: Change in the Share of Private Rental Housing across Income Percentiles

Notes: For each household income percentile, we first calculate the share of properties in the private rental sector in 2011 and 2017. The figure reports the difference between the 2017 and 2011 shares for each income percentile.

way fixed effect specification:

$$Private_rent_{it} = \alpha + \beta Violate_{it} + \gamma_i + \phi_t + \epsilon_{it}. \tag{17}$$

In this equation, $Private_rent_{it}$ is a dummy variable indicating whether house i is a private rental house in year t. We define $Private_rent_{it}$ as 100 if the house i is a privately rented house in year t, and 0 otherwise. $Violate_{it}$ is a dummy variable to indicate whether the $MPTI_i$ of house i is larger than $NIBUD_t$ in year t. If $MPTI_i$ is larger than $NIBUD_t$, we define $Violate_{it} = 1$, otherwise, 0. γ_i indicates property fixed effects and ϕ_t indicates year fixed effects, which capture the time-invariant property characteristics and common trend of properties over the year, respectively. ϵ_{it} indicates the error term. We have no time-variant covariates at the property level, but in some specifications, we control for property type (apartment or single-family house) multiplied by a linear year trend.[23] Because Glaeser and Gyourko (2007) finds that rental homes in the US are more likely to be apartments, while owner-occupied homes are more likely to be single-family houses, we thus allow for different time trends in ownership status for apartments and single-family houses.

In all specifications, we include property and year fixed effects, as indicated in the estimating equation. To account for potential local trends in buy-to-let activity, we also estimate specifications that include postcode $\times$ year fixed effects.[24] Controlling for time-varying local factors at such a fine geographical scale is particularly useful because buy-to-let investment activity is known to vary substantially within cities.

Dynamic treatment effects. The key assumption for identifying ATT is that the parallel trend assumption holds, which allows us to interpret the control group as a counterfactual of the treatment group. The parallel trend assumption is untestable, but we can build some confidence that it holds if we observe a parallel pre-trend of the treatment and control group. We conducted an event study to check the parallel trend and to investigate the dynamic effects of the borrowing constraint. We carried out event studies by estimating:

$$Private_rent_{it} = \alpha + \sum_{\tau=-5, \tau\neq-1}^{\tau=5} \beta_\tau \mathbb{I}(t - T_i = \tau) + \gamma_i + \rho_p \phi_t + \epsilon_{it}. \tag{18}$$

T_i denotes the year in which house i first violated the NIBUD rule. $\rho_p \phi_t$ denotes the 4-digit postcode times year fixed effect.

[23] We also experimented with controlling for nonlinear time trends, but the estimates are almost the same as those under linear trends. To maintain brevity, we do not present those.

[24] We use four-digit postcode areas, defined by the first four digits of the Dutch postcode. The 4,770 areas defined in this way cover the entire Netherlands and typically consist of a small number of contiguous streets.

Treatment effects heterogeneity. We adopt prevailing "heterogeneity-robust" diagnostics for staggered treatment timing (In our case, the house could start to violate the NIBUD rule in any year between 2011 and 2017). A recent econometric literature pointed out that the two-way fixed effect estimator can be interpreted as a weighted average estimator of each staggered treatment effect (de Chaisemartin and D'Haultfœuille, 2020; Goodman-Bacon, 2021). If there were heterogeneity of treatment effects over time and units, we might get negative weights which make the TWFE estimator hard to interpret.

We calculate and plot the weights of each treatment following de Chaisemartin and D'Haultfœuille (2020). Figure A4 shows that all weights associated with each treatment are non-negative and there is little variation. Moreover, we adopt the heterogeneity-robust estimator of Callaway and Sant'Anna (2021) and the standard TWFE estimator as the main estimators. We also present the estimation from Borusyak et al. (2024) and Sun and Abraham (2021) in the appendix.

Results discussion. Table 1 presents the results of our baseline estimation. We present the dynamic treatment effects of 4 estimators in Figure 12. These results capture the average treatment effects on the treated (ATT) of the violation of the NIBUD rule. We find the expected positive and significant coefficients in all specifications. The estimators show that houses subject to borrowing constraints have on average a 0.77% to 1.32% higher probability of becoming a private rental property than houses that do not violate the NIBUD. We conclude that our empirical analysis confirms a key result of our assignment model: tighter borrowing constraints open up profitable arbitrage possibilities for buy-to-let (or keep-to-let) investors in those parts of the market where demand is depressed.

We then discuss the magnitude of the treatment effects starting with the sample mean. In the treatment group, the average probability of a house being privately rented over the years was 4.03%, initially 0.61%, and reaching 8.86% by the end of 2017. Our estimates show that the probability of a house entering the private rental sector increases by 0.8% to 1.3%, which is a large figure relative to the sample mean. If compared to the sample mean in a naive way, our estimated results can explain the growth in the private rental market share by 19% to 33% using violation of the NIBUD rule.

The coefficient plots show that there are no significant differences between the treatment and control groups in the private rented sector before violating the NIBUD rule. The parallel pre-trend makes us trust that the parallel trend assumption is likely to hold. After violating the NIBUD rule, however, properties subject to borrowing restrictions start to have a significantly greater probability of being converted to the private rental house. We also see cumulative treatment effects over the period. After a house violates NIBUD rules for the first time, the probability of it becoming a private rent becomes greater and slowly increases.

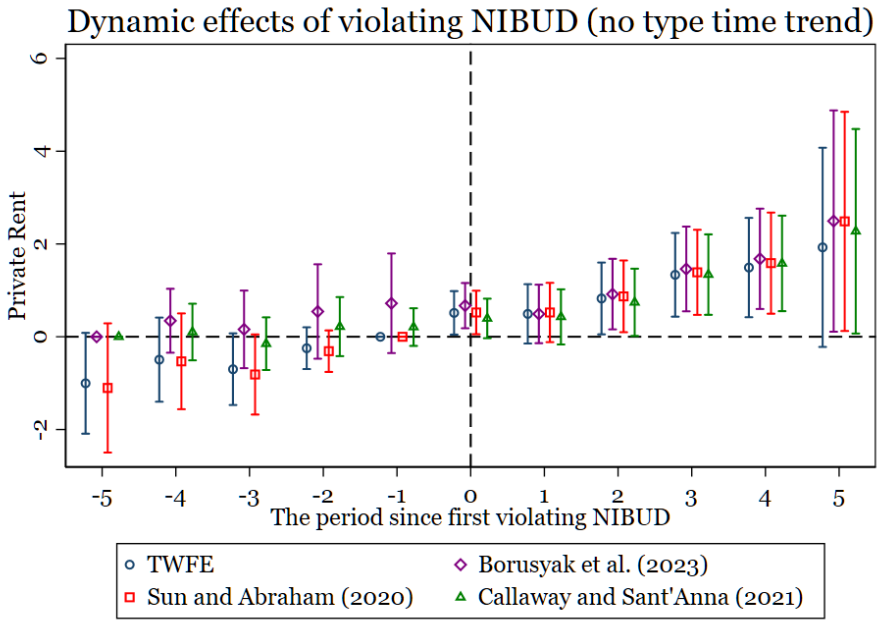
We further decompose the treatment effects by group of year entering treatment and by calendar years, presented in Figure A5. The results show that the treatment effects are almost positive for all groups; however, only significant in the groups 2012 and 2013. It suggests that the estimated treatment effects are dominated by the sample entering the treatment group in 2012 and 2013. The results also show that there are cumulative effects; by the end of 2017, the cumulative treatment effects range from 1.23% to 1.41%. In other words, by the end of 2017, the private rental sector share in the treatment group sample was 8.86%, of which 1.23% to 1.41% was contributed by borrowing constraints, accounting for approximately 14% to 16%. This suggests that borrowing constraints have strong explanatory power for the growth of the private rented sector. In addition, with the linear year trend, it is intuitive that apartments have a greater probability of becoming private rental housing.

This result is likely to be an underestimate. On one hand, the instability of MPTI could be a key factor. If MPTI fluctuates significantly over time, it could lead to misclassification errors for samples near the borderline of borrowing constraints. This would lead to samples that should belong to the treatment group being incorrectly placed in the control group, while samples that should be in the control group are mistakenly placed in the treatment group, thereby reducing the estimate. On the other hand, the sample we used consists of properties that were recently transacted in 2009 and 2010, and these are unlikely to be re-transacted between 2011 and 2017. The smaller house tenure turnover rate could cause the estimates to be biased downward. We performed a simple calculation and found that the turnover rate of our sample is approximately 1.4 times that of the overall turnover rate for all properties that were ever transacted during 2011 and 2017. This implies that we can blow up our estimates, allowing borrowing constraints to explain the growth in the private rental housing market by 20% to 22%.

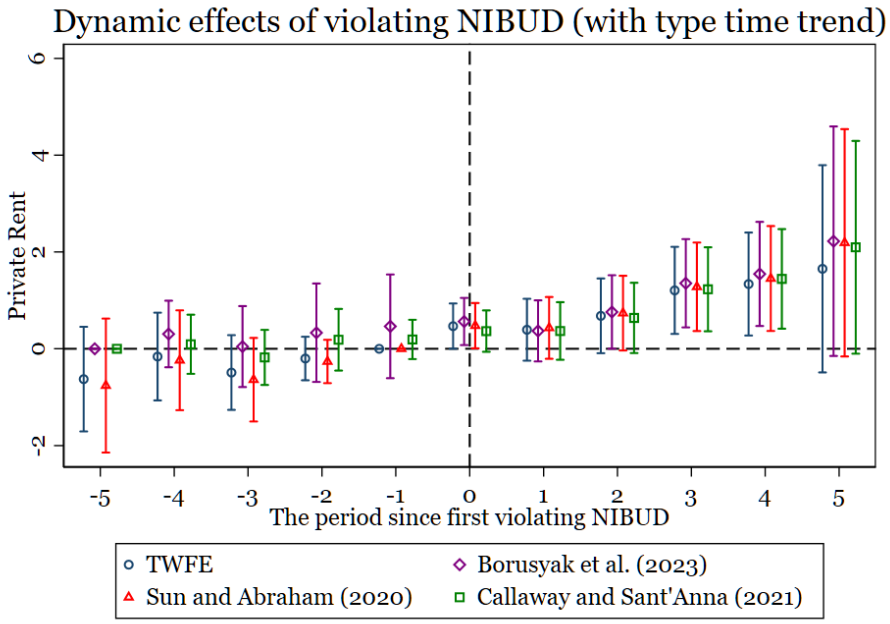
We will further discuss how to adjust MPTI to obtain a more accurate measure of whether a property is constrained and to improve the estimation results.

4.2 Wealth heterogeneity

Our assumption is that the MPTI of the same house across years remains constant, as bought by the household with the same income, but this assumption may be violated due to heterogeneity in households or properties. However, there may be significant wealth heterogeneity among homebuyers in our sample. Calculating the MPTI based on their loan amounts and assuming the same MPTI in subsequent years overlooks this wealth heterogeneity. We further consider household wealth heterogeneity by allowing households to use non-housing assets to either reduce or increase their total loan amount, resulting in different MPTIs. This means that we assume a homebuyer with the same income but zero wealth in subsequent years would purchase the property.



(a) Without type-specific linear time trends



(b) With type-specific linear time trends

Figure 12: Dynamic Effects of Borrowing Constraints

Notes: This figure presents dynamic treatment effect estimates with 95% confidence intervals based on equation (18). Panel (a) reports estimates without type-specific linear time trends, while Panel (b) includes them, corresponding to the specification in Columns (2) and (4) of Table 1, respectively. To account for treatment effect heterogeneity under staggered treatment adoption, we report estimates from four estimators: two-way fixed effects (TWFE), Borusyak et al. (2024), Sun and Abraham (2021), and Callaway and Sant'Anna (2021). Standard errors are clustered at the property level.

Table 1: Baseline Difference-in-Differences Estimates

	Private Rental							
	TWFE				CSDID			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Violation	1.320***	0.998***	0.963***	0.833***	1.175***	0.859***	0.936***	0.774***
	(0.251)	(0.259)	(0.248)	(0.258)	(0.256)	(0.249)	(0.255)	(0.250)
Type × Year trend			1.652***	1.386***				
			(0.052)	(0.070)				
N	346,470	343,336	346,470	343,336	346,162	346,162	346,162	346,162
R^2	0.449	0.480	0.456	0.482				
Type × Year trend			✓	✓			✓	✓
Property FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
PC4 × Year FE		✓		✓		✓		✓

Notes: The table reports estimates from two-way fixed-effects (TWFE) regressions and the estimator of Callaway and Sant'Anna (2021). The dependent variable is an indicator for whether a property is privately rented. *Violation* is an indicator equal to one if the property's mortgage-payment-to-income ratio (MPTI) in a given year exceeds the maximum threshold prescribed by the NIBUD rules, and zero otherwise. Property type distinguishes between single-family houses and apartments. PC4 denotes the four-digit postal code area. Standard errors are clustered at the property level. All specifications include property and year fixed effects. Columns (2), (4), (6), and (8) include PC4-by-year fixed effects, while Columns (3), (4), (7), and (8) additionally control for type-specific linear time trends. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

We report this wealth heterogeneity correction result in Table 2 and Figure 13. After correcting for household wealth heterogeneity, we obtain consistent and larger estimates. This suggests that correcting for household wealth heterogeneity helps reduce the misclassification bias in determining whether a house belongs to the treatment or control group, leading to larger estimates. Overall, adjusting loans and MPTI based on the wealth of existing homebuyers makes these households more representative, which in turn makes it more likely that MPTI will exhibit stability across years. Our estimated results and dynamic treatment effects validate the reliability of this assumption and the robustness of the baseline results.

4.3 Macroeconomic trends

Our identification strategy relies on the assumption of MPTI stability. Although we have verified that the MPTI of the same house remains relatively stable across years, certain macroeconomic factors may still affect its stability. We further account for changes in household income. We adopt a methodology of getting the counterfactual MPTI which is conceptually aligned with Bosshardt et al. (2024). Our procedure is as follows: First, we establish a linear prediction between homebuyers' MPTI and household income based on repeated transactions from 2006 to 2010 with house and year fixed effects, based on specification (19). The underlying intuition is that for the same property, there exists a certain relationship between different buyers' incomes and their MPTI, which is indicated by the coefficient β. Next, we collect household income data for those who moved into the baseline sample (houses transacted in 2009 and 2010) between 2011 and 2017 (including homebuyers) and compute the annual income trend changes. Using this annual trend and β, we get the predicted MPTI for households from 2011 to 2017 based on (20). Finally, we determine whether a house is constrained based on the adjusted predicted MPTI and the NIBUD rule, and we repeat the baseline regression using this new variable.

$$MPTI_{it} = \alpha + \beta Income_{it} + \eta_i + \lambda_t + \epsilon_{it} \tag{19}$$

$$Pre_MPTI_{it} = \beta(\frac{1}{N_t}\sum_{i=1}^{N_t} Income_resident_{it} - Income_i) + MPTI_i \tag{20}$$

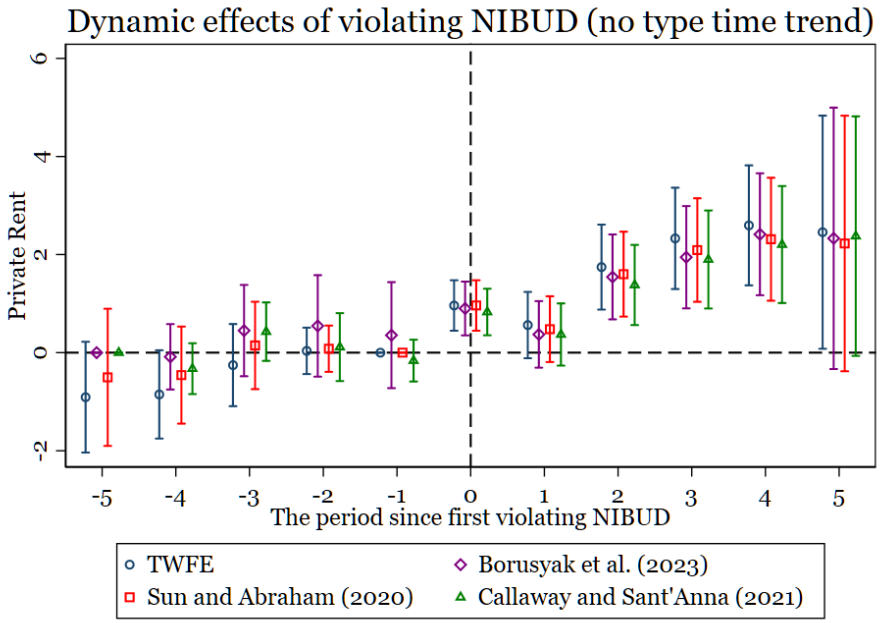
In which, $Income_resident_{it}$ denotes the new resident i's household income in year t. $Income_i$ and $MPTI_i$ denote the homebuyer's household income and MPTI for house i in 2009 and 2010.

We present the results in Table 3 and Figure 14. Due to a significant decline in the income of new movers from 2011 to 2013, a large portion of the corrected first-time borrowing constraints occurred in these years, while the sample with first-time borrowing constraints became much smaller in 2015–2017. As a result, in the event study, the standard errors of the coefficients representing the pre-trend

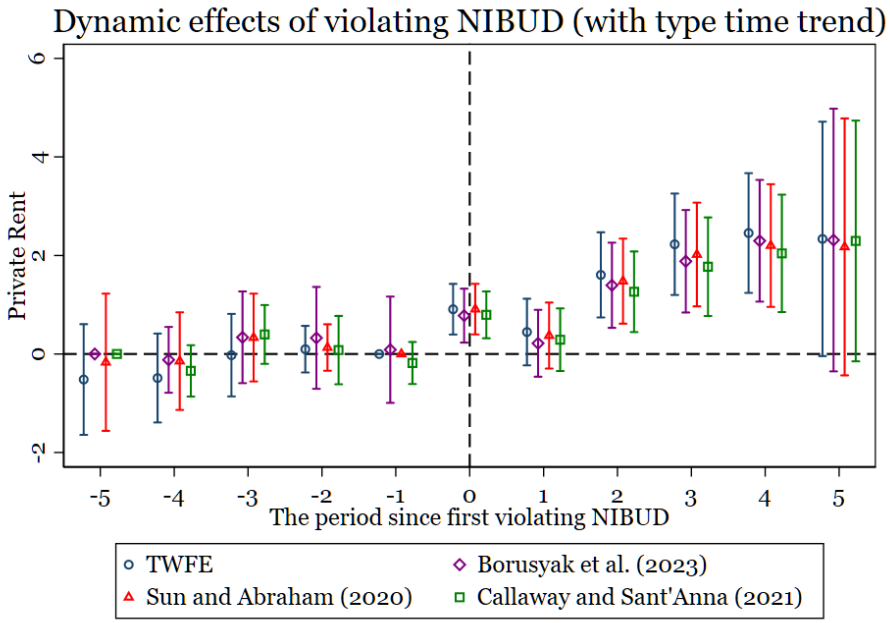
Table 2: Correction for Household Wealth Heterogeneity

	Private Rental							
	TWFE				CSDID			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Violation	1.825***	1.454***	1.403***	1.275***	1.680***	1.251***	1.417***	1.157***
	(0.285)	(0.291)	(0.281)	(0.290)	(0.285)	(0.275)	(0.283)	(0.276)
Type × Year trend			1.636***	1.380***				
			(0.050)	(0.068)				
N	364,186	361,115	364,186	361,115	363,952	363,952	363,952	363,952
R^2	0.449	0.479	0.456	0.481				
Type × Year trend			✓	✓			✓	✓
Property FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
PC4 × Year FE		✓		✓		✓		✓

Notes: The table reports estimates from two-way fixed-effects (TWFE) regressions and the estimator of Callaway and Sant'Anna (2021), correcting for household wealth heterogeneity. Variables are defined as in Table 1. Standard errors are clustered at the property level. All specifications include property and year fixed effects. Columns (2), (4), (6), and (8) include PC4-by-year fixed effects, while Columns (3), (4), (7), and (8) additionally control for type-specific linear time trends. Alternative estimation results are reported in Table A4. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.



(a) Without type-specific linear time trends



(b) With type-specific linear time trends

Figure 13: Dynamic Effects of Borrowing Constraints with Wealth Heterogeneity Correction

Notes: This figure presents dynamic treatment effect estimates with 95% confidence intervals after accounting for wealth heterogeneity. Panel (a) reports estimates without type-specific linear time trends, while Panel (b) includes them, corresponding to the specification in Columns (2) and (4) of Table 2, respectively. To account for treatment effect heterogeneity under staggered treatment adoption, we report estimates from four estimators: two-way fixed effects (TWFE), Borusyak et al. (2024), Sun and Abraham (2021), and Callaway and Sant'Anna (2021). Standard errors are clustered at the property level.

become very large. However, the corrected violation of borrowing constraints still has a significantly positive effect on whether a house becomes a private rental property, which again confirms our idea.

4.4 Robustness checks

Sample extension. The baseline regression uses a sample of houses transacted in 2009 and 2010, the two years immediately preceding the NIBUD tightening. However, these transactions may be less likely to occur again between 2011 and 2017, potentially limiting representativeness. On the other hand, expanding the sample to include transactions before 2009 may challenge the assumption of MPTI stability. We face a trade-off between MPTI stability and sample representativeness when selecting the sample. To confirm the robustness of the baseline results, we extend the transaction sample to include 2006–2010 as a robustness check. This increases the likelihood that repeatedly transacted houses are converted into private rental properties, leading to larger estimated coefficients. However, on the other hand, sacrificing MPTI stability may result in misclassification of the treatment and control group, potentially biasing the estimated coefficients downward. Overall, whether the estimator becomes larger or smaller is uncertain.

We exhibit the results in Table A6 and Figure A7. They show a consistent, while smaller, effect of borrowing constraints on private renting. This may be due to the relatively poor stability of MPTI in the 2006-2008 transaction sample between 2011 and 2017, which introduced more misclassification bias, thereby dominating the estimated results and leading to smaller estimates.

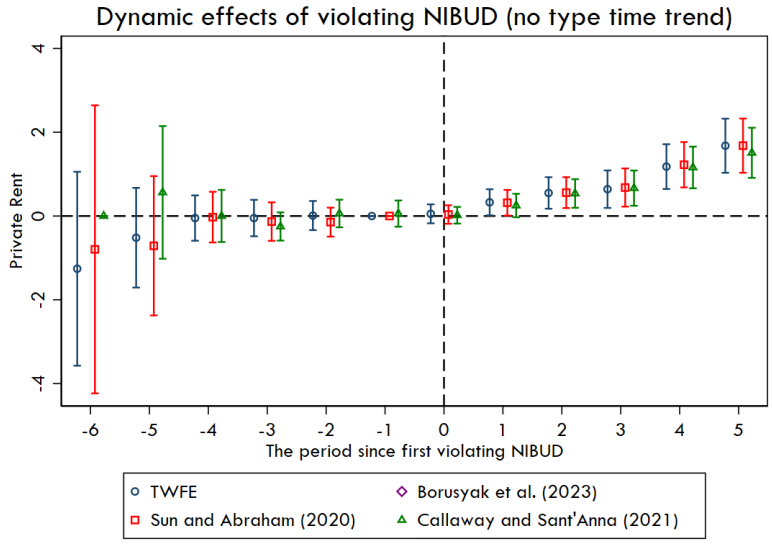
Placebo test. One concern regarding the empirical results is that the observed treatment effects may be driven by some unknown random factors. To address this, we conduct a placebo test. In this placebo test, we randomly generated a fictitious treatment group within the original sample for 1000 times. The size of the fictitious group matches that of the actual treatment group, and the distribution of their treatment timing is identical to that of the real treatment group. This approach allows us to randomly assign each observation to the treatment group, enabling us to examine whether such a random distribution would produce similar treatment effects.

We plot the estimators and p-value of these 1000 estimations in the Figure A8. These estimators exhibit a distribution resembling a normal distribution. The actual treatment effect of 0.833 is larger than all the fictitious estimators we got. We hereby reject the hypothesis that the real treatment effect is equal to the fictitious treatment effect. We rule out the possibility that the treatment effect is driven by randomness.

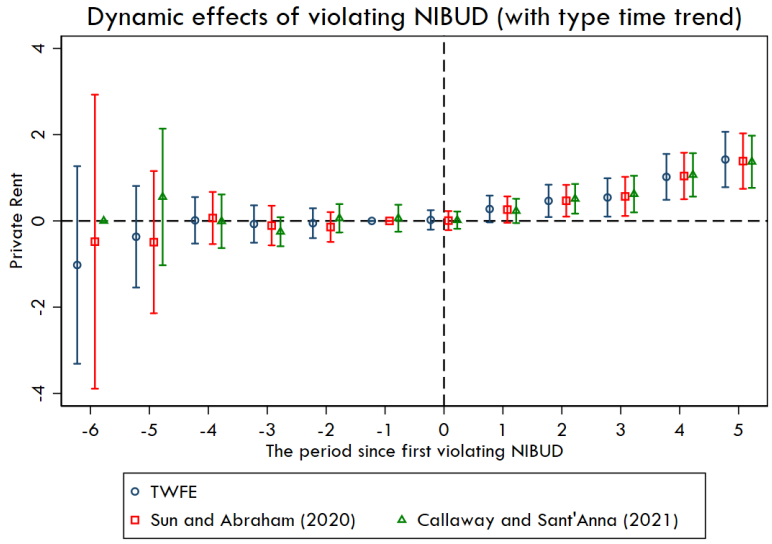
Table 3: Correction for Household Income Trends

	Private Rental							
	TWFE				CSDID			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Violation	0.406***	0.376***	0.369***	0.326**	0.786***	0.591***	0.659***	0.548***
	(0.137)	(0.143)	(0.135)	(0.142)	(0.136)	(0.131)	(0.136)	(0.132)
Type × Year trend			1.644***	1.406***				
			(0.054)	(0.073)				
N	317,484	314,429	317,484	314,429	316,319	316,360	316,319	316,360
R^2	0.447	0.480	0.454	0.482				
Type × Year trend			✓	✓			✓	✓
Property FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
PC4 × Year FE		✓		✓		✓		✓

Notes: The table reports estimates from two-way fixed-effects (TWFE) regressions and the estimator of Callaway and Sant'Anna (2021), correcting for household income trends. Variables are defined as in Table 1. Standard errors are clustered at the property level. All specifications include property and year fixed effects. Columns (2), (4), (6), and (8) include PC4-by-year fixed effects, while Columns (3), (4), (7), and (8) additionally control for type-specific linear time trends. Alternative estimation results are reported in Table A5. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.



(a) Without type-specific linear time trends



(b) With type-specific linear time trends

Figure 14: Dynamic Effects after Adjusting for Household Income Trends

Notes: This figure presents dynamic treatment effect estimates with 95% confidence intervals after adjusting for household income trends. Panel (a) reports estimates without type-specific linear time trends, while Panel (b) includes them, corresponding to the specifications in Columns (2) and (4) of Table 3, respectively. To account for treatment effect heterogeneity under staggered treatment adoption, we report estimates from three estimators: two-way fixed effects (TWFE), Sun and Abraham (2021), and Callaway and Sant'Anna (2021). Standard errors are clustered at the property level. We do not report estimates from Borusyak et al. (2024) because, after additionally absorbing PC4 $\times$ year fixed effects, the limited size of the treatment group leaves insufficient identifying variation for the *did_imputation* procedure to produce estimates. The relatively large standard errors for event times -6 to -4 reflect the very small number of properties that first enter treatment in 2015–2017.

4.5 Matching sample

In the previous subsection, we always restrict the analysis to houses transacted just before the change in the Code of Conduct of 2011. However, it may be argued that this is an idiosyncratic set of houses; for example, it may be because dwellings that are frequently traded are over-represented. This may cause these houses to be more prone to trades or change ownership status, thus biasing our estimates. To address this issue, we reconstruct the sample by using properties similar to these transaction samples. That is, we assume that houses with characteristics that are similar to those in the sample used thus far also have a similar probability of becoming treated in the period 2012-2017. We use location, type, price, and floor area to construct this matched sample (e.g. other apartments in the same condominium) and continue to use the MPTI of the property in the original sample to determine whether the new property violates NIBUD. Our hypothesis is that the ownership status of the matched property is as likely to switch from owner-occupied to private rental as that of the original property.

We conduct the following matching procedure. First, for each baseline property, we identify the full set of candidate properties with the same six-digit postcode and property type. This defines our full matching sample. We then impose an additional restriction on assessed property values, requiring the absolute difference in assessed value between the baseline and matched properties to be less than €1,000. This more restrictive matching criterion improves comparability between baseline and matched properties, at the cost of a smaller matched sample.

The estimates based on the two matching samples are reported in Table 4 and Figure 15. Panel A presents the results using all properties of the same type located within the same six-digit postcode area as the corresponding baseline property. The results show that properties with similar location and housing-type characteristics also exhibit an increased likelihood of transitioning from owner occupation to private rental when the corresponding baseline property is subject to the constraint. The estimated effects are smaller than those in the baseline specification, but remain highly robust and statistically significant. The attenuation in the estimated coefficients is likely to reflect measurement error arising from assigning the violation status of the baseline property to its matched properties. At the same time, matched properties may have higher underlying rates of tenure turnover, which would tend to increase the likelihood of observing a transition. The fact that the estimated effects are nevertheless smaller suggests that the attenuation induced by treatment misclassification dominates this countervailing force.

Panel B further restricts the matched sample to properties whose assessed values differ from those of the corresponding baseline properties by less than €1,000. With this tighter matching criterion, the estimated treatment effects become larger and approach the baseline estimates, although they remain slightly smaller. The resulting reduction in sample size, however, lowers statistical precision. Overall,

the matched-sample results show that our findings are not driven by the specific composition of the baseline transaction sample, but reflect a broader increase in private rental housing associated with credit constraints across the Dutch housing market.

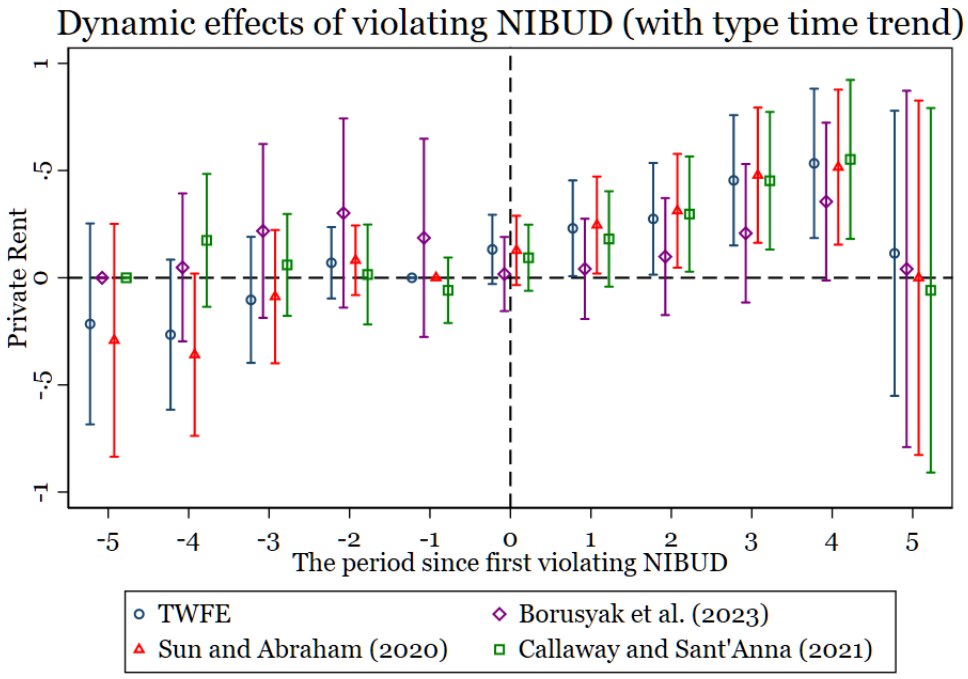


Figure 15: Dynamic Effects in the Matched Sample

Notes: This figure presents dynamic treatment effect estimates with 95% confidence intervals using the matched sample, corresponding to the specifications in Columns (4) and (8) of Panel A of Table 4. The matched sample includes all properties with the same six-digit postcode and property type as the corresponding baseline property. Treatment status is assigned based on the MPTI constraint status of the corresponding baseline property. Standard errors are clustered at the property level.

4.6 Price effect

We consider whether house transaction prices and transaction volume are affected by borrowing constraints. Since there is a very small sample of houses that were transacted in 2009 and 2010 and then transacted again between 2011 and 2017, we cannot apply the same specification as the baseline regression. Alternatively, we calculate the average house violation probability at each Pc4 level from 2011 to 2017 using the violation status of houses in 2009 and 2010 and the decreasing NIBUD rule. The results presented in Table 5 show that the borrowing constraints at the Pc4 level cannot explain the local housing transaction prices and volumes. This suggests that buy-to-let investors (or keep-to-let investors) have full arbitrage. Investors enter the market and fill the gap in prices caused by credit-constrained buyers leaving the market. Consistent with the specification of this empirical part,

Table 4: Matching Sample Regression

	TWFE				CSDID			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Full Matched Sample								
Violation	0.663***	0.397***	0.438***	0.267***	0.517***	0.328***	0.368***	0.273***
	(0.096)	(0.096)	(0.096)	(0.096)	(0.100)	(0.100)	(0.100)	(0.100)
Type × Year trend			0.883***	0.992***				
			(0.025)	(0.031)				
N	3,938,103	3,937,881	3,938,103	3,937,881	3,926,297	3,926,297	3,926,297	3,926,297
R^2	0.732	0.736	0.733	0.736				
Panel B: Matched Sample with Assessed Value Difference ≤ 1,000								
Violation	0.968***	0.652	0.692*	0.533	0.717*	0.564	0.659*	0.530
	(0.367)	(0.401)	(0.365)	(0.400)	(0.371)	(0.367)	(0.369)	(0.367)
Type × Year trend			1.275***	1.219***				
			(0.071)	(0.109)				
N	258,708	256,237	258,708	256,237	257,995	257,995	257,995	257,995
R^2	0.734	0.750	0.735	0.750				
Type × Year trend			✓	✓			✓	✓
Property FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
PC4 × Year FE		✓		✓		✓		✓

Notes: The table reports estimates from two-way fixed-effects (TWFE) regressions and the estimator of Callaway and Sant'Anna (2021) using matched samples. Variables are defined as in Table 1. The matched sample consists of properties with the same PC6 and property type as properties in the baseline sample. Panel A uses the full matched sample, while Panel B restricts the matched sample to properties whose assessed value differs from that of the corresponding baseline property by no more than 1,000. Violation status is inherited from the corresponding property in the baseline sample, while private rental status is measured for the matched property. Standard errors are clustered at the property level. All specifications include property and year fixed effects. Columns (2), (4), (6), and (8) include PC4-by-year fixed effects, while Columns (3), (4), (7), and (8) additionally control for type-specific linear time trends. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

we also regress a dummy variable for whether the house is privately rented on the average violation probability at the Pc4 level (See it in Table A8). We find that the local average violation probability still has a significant positive impact on whether a house is privately rented, consistent with our baseline results.

The combined evidence suggests that investors enter the market to fully arbitrage, converting credit-constrained housing into private rental housing without significant impacts on user costs (house prices). This is consistent with the predictions of our model.

4.7 Buy-to-let or keep-to-let

So far, we have shown how borrowing constraints affect the likelihood that properties transition into the private rental sector. We next examine whether this expansion is driven primarily by buy-to-let (BTL) investment or by keep-to-let (KTL) behavior. To distinguish between the two, we link changes in property tenure status to the transaction records. We classify a property as BTL if it is privately rented in a given year and has been transacted between July 2011 and the end of that year, consistent with a purchase followed by rental. We classify it as KTL if it is privately rented but has no transaction record over the same period, consistent with an existing owner retaining the property and renting it out.

The estimators using BTL and KTL respectively as the explained variables are shown in the Table A7. The results indicate that violating the NIBUD rule resulted in significant and similar positive effects for BTL and KTL, although the effect of borrowing constraints on KTL is somewhat larger. This may be due to the lower transaction probability in the baseline sample during 2012-2017.

5 Conclusion

This paper provides an analysis of the effect of borrowing constraints on buy-to-let investment where the housing stock can be considered as given, at least in the short run. Binding borrowing restrictions open up possibilities for profitable arbitrage by buy-to-let investors. By offering the houses preferred by the restricted households as rental housing, they allow them to reach the same level of housing consumption as in the case without borrowing constraints, albeit at a higher price than is relevant in the situation when borrowing constraints are binding. In the equilibrium with free entry of buy-to-let investors, the allocation of housing over households is the same as in the situation without borrowing constraints.

We use this assignment model to predict which owner-occupied houses will be converted to rental houses, and test these predictions exploiting a series of contractions of the debt-service-to-income constraint in the Netherlands between 2012 and 2016. The empirical analysis showed that in the Netherlands switches from the

Table 5: Price Effects of Borrowing Constraints

	Log Sale Price	
	(1)	(2)
PC4 Violation Rate	-0.00755	-0.00730
	(0.0140)	(0.0109)
Property Type	-0.116***	-0.116***
	(0.00566)	(0.00560)
Log Size	0.700***	0.701***
	(0.00965)	(0.00936)
N	1,730,079	1,730,079
R^2	0.669	0.675
Age Interval FE	✓	✓
Year FE	✓	✓
PC4 FE	✓	✓
Municipality × Year FE		✓

Notes: The table examines whether borrowing constraints at the PC4-year level affect transaction prices. The dependent variable is the log transaction price. PC4 Violation Rate is the average violation probability at the PC4-year level, constructed by combining the MPTI of baseline transactions in 2009 and 2010 with the NIBUD rules in effect from 2011 to 2017. The sample includes all transactions from 2011 to 2017 in PC4 areas covered by the baseline sample. All specifications include age-interval, year, and PC4 fixed effects, while Column (2) additionally includes municipality-by-year fixed effects. Standard errors are clustered at the property level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

owner-occupied to private rental sector were significantly more prevalent among houses that had been involved in transactions that became infeasible after the tightening of the borrowing constraints. The series of contractions that ultimately affected 17 percent of households resulted in a cumulative increase of the private rental sector of 1.2 to 1.3 percentage points, accounting for 14 to 16 percent of the overall increase in this sector by the end of 2017. However, our estimates suffer from significant attenuation bias and may be lower because we underestimate the turnover rate. Correcting these biases, the introduction and subsequent tightening of mortgage payment-to-income constraints may explain up to 75 percent of the total increase in the private rental sector.

These findings show that the borrowing constraints that have been introduced in many countries in the wake of the financial crisis have likely significantly contributed to the observed rise of the private rental sectors in many developed economies.

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Appendix

A Additional theoretical results

A.1 Proofs

This appendix establishes two supporting results that underlie the analysis of the equilibrium without borrowing constraints and presents the proofs of Lemmas 1 and 2 and Proposition 1.

Lemma 3 (Monotonicity of housing consumption)**.** *Households with a higher income y consume weakly more housing services q.*

Proof. Consider two households, indexed by 1 and 2, with incomes $y_1 > y_2$. Since household 1 has a strictly larger budget set and preferences are increasing, it attains a strictly higher utility level in equilibrium. Let these utility levels be denoted by u_1 and u_2, and suppose, contrary to the claim, that the households consume housing quantities $q_1 < q_2$.

Let $c(q, u)$ denote the level of other consumption that, together with housing consumption q, yields utility u. Since $u_1 > u_2$, it follows that

$$c(q, u_1) > c(q, u_2)$$

for all q. The willingness to pay of household 1 for moving from q_1 to q_2, while remaining on utility level u_1, is

$$WTP = \int_{q_1}^{q_2} M(q, c(q, u_1))\, dq.$$

The willingness to accept of household 2 for moving from q_2 to q_1, while remaining on utility level u_2, is

$$WTA = \int_{q_1}^{q_2} M(q, c(q, u_2))\, dq.$$

Because $M(q, c)$ is increasing in c for a given q by the single-crossing assumption, we have

$$M(q, c(q, u_1)) > M(q, c(q, u_2))$$

for all $q \in [q_1, q_2]$. Hence, $WTP > WTA$, so there exists a transfer at which the two households would both prefer to exchange houses. This contradicts equilibrium. Therefore, a higher-income household cannot consume less housing than a lower-income household. □

We now prove Lemma 1.

Proof of Lemma 1. We proceed in two steps.

First, we show that the user cost function $p(q)$ is strictly increasing and continuous in q. Suppose, to the contrary, that there exist $q' > q$ such that $p(q') \leq p(q)$.

Then any household that can afford the lower-quality house q would weakly prefer q' and, because q' is of strictly higher quality, would strictly prefer it, implying zero demand for q. This contradicts equilibrium. Hence, $p(q)$ must be strictly increasing.

Next, suppose that $p(q)$ has a discontinuity at some q. Then the marginal price of housing becomes arbitrarily large at that point, implying zero demand for qualities just above the discontinuity. This again contradicts market clearing. Therefore, $p(q)$ must be continuous.

Lemma 3 establishes that higher-income households cannot consume less housing than lower-income households. Combined with the monotonicity of $p(q)$, this implies that the equilibrium allocation is assortative: higher-quality houses are assigned to higher-income households.

Market clearing then requires that the S highest-income households are matched to the S available houses, which yields (9). Continuity of the assignment follows from differentiating (9), which gives

$$\frac{dy}{dq} = \frac{g(q)}{f(y)}, \tag{A1}$$

where $g(q)$ and $f(y)$ are continuous and strictly positive. □

We next show that, along the equilibrium assignment, the user cost schedule is locally less convex than the household's indifference curve. This curvature condition ensures that the local optimality condition used to derive Lemma 2 is sufficient for optimality.

Lemma 4 (Sufficiency of local optimality)**.** *Let $\pi(q) \equiv dp/dq$ denote the marginal user cost of housing. Along the equilibrium assignment,*

$$\frac{d\pi}{dq} > \left.\frac{dM}{dq}\right|_{u \text{ constant}}. \tag{A2}$$

Proof. In equilibrium, $\pi(q) = M(q, c)$ with $c = y(q) - p(q)$. Taking total derivatives,

$$\frac{d\pi}{dq} = \frac{\partial M}{\partial q} + \frac{\partial M}{\partial c}\left(\frac{g(q)}{f(y)} - \pi\right),$$

where we use $dc = d(y(q) - p(q)) = \left(\frac{g(q)}{f(y)} - \pi\right) dq$ and the assignment rule.

Along an indifference curve, $dc = -M dq$, so

$$\left.\frac{dM}{dq}\right|_{u \text{ constant}} = \frac{\partial M}{\partial q} - \pi \frac{\partial M}{\partial c}.$$

Subtracting yields

$$\frac{d\pi}{dq} - \left.\frac{dM}{dq}\right|_{u \text{ constant}} = \frac{\partial M}{\partial c}\frac{g(q)}{f(y)} > 0,$$

since $\partial M/\partial c > 0$ by the single-crossing assumption and $g(q), f(y) > 0$. This implies (A2). □

We are now ready to prove Lemma 2.

Proof of Lemma 2. Consider a household with income $y(q)$ that is assigned a house of quality $q \in (q^{\min}, q^{\max})$. Taking the user cost schedule $p(\cdot)$ as given, the household solves

$$\max_{q'} \; u(q', y(q) - p(q')).$$

At an interior optimum, the first-order condition is

$$\frac{\partial u}{\partial q}(q', c) - \frac{\partial u}{\partial c}(q', c)\frac{dp}{dq'} = 0,$$

where $c = y(q) - p(q')$. Evaluating this condition at the assigned quality $q' = q$ yields

$$\frac{dp}{dq} = \frac{\partial u/\partial q}{\partial u/\partial c} = M(q, y(q) - p(q)),$$

which establishes (10).

The boundary condition follows from the requirement that the marginal participant with income y^P is indifferent between the lowest-quality house and the outside option, which determines $p(q^{\min})$ through (8).

By Lemma 4, the user cost schedule is locally less convex than the indifference curve at the optimum. Hence the first-order condition characterizes the household optimum. Therefore, any continuously differentiable function satisfying (10) and the boundary condition supports the equilibrium assignment.

Finally, under the maintained assumptions that u is C^2 with $u_c > 0$, the function $M(q, c)$ is continuous in q and locally Lipschitz in c on the relevant domain. Together with the continuity of $y(q)$, the differential equation with boundary condition $p(q^{\min})$ admits a unique solution on $[q^{\min}, q^{\max}]$. Hence, the equilibrium user cost function is unique. □

We now prove Proposition 1.

Proof of Proposition 1. The proof proceeds by constructing the constrained user cost schedule along the fixed assignment rule $q(y)$ and then verifying that this assignment is supported in equilibrium.

If no income y^* exists, the unconstrained user cost schedule satisfies the borrowing constraint for every household. The borrowing constraint is therefore slack throughout the unconstrained equilibrium, so the constrained and unconstrained equilibria coincide.

Suppose instead that y^* exists. Since the borrowing constraint is slack on $[y^P, y^*]$, the user cost schedule supporting the assignment $q(y)$ is characterized by Lemma 2:

$$\frac{dp(q(y))}{dy} = M(q(y), y - p(q(y)))\frac{f(y)}{g(q(y))}.$$

The constrained and unconstrained schedules have the same initial condition at y^P. By uniqueness of the solution to this differential equation, they therefore coincide up to y^*.

At y^*, the unconstrained schedule would violate the borrowing constraint. The constrained schedule thus cannot continue to satisfy the unconstrained equilibrium condition. Instead, the borrowing constraint binds, so that

$$p^C(q(y)) = \mu y.$$

Equivalently,

$$\frac{\partial p^C}{\partial q} = \mu \frac{g(q(y))}{f(y)}.$$

The constraint continues to bind as long as, evaluated at $p^C(q(y)) = \mu y$,

$$M(q(y), (1-\mu)y) \frac{f(y)}{g(q(y))} \geq \mu.$$

On this interval, households' marginal willingness to pay for housing is at least as large as the marginal price implied by the borrowing constraint.

If this inequality is violated at some income y'', the slope implied by the borrowing constraint exceeds households' marginal willingness to pay. The constraint then becomes slack, and the constrained schedule again satisfies

$$\frac{dp^C(q(y))}{dy} = M(q(y), y - p^C(q(y))) \frac{f(y)}{g(q(y))}$$

with initial condition

$$p^C(q(y'')) = \mu y''.$$

If the resulting schedule reaches the borrowing constraint again, the same argument applies, yielding the recursive construction.

It remains to verify that the assignment rule $q(y)$ is supported. On intervals where the borrowing constraint is slack, the user cost schedule is constructed along the assignment $q(y)$ in the same way as in the unconstrained equilibrium. Hence households are locally optimized at their assigned housing qualities. By the single-crossing property, this local optimality implies that no household can profitably deviate to another housing quality on these intervals.

On intervals where the borrowing constraint binds, households would like to consume higher-quality housing, but cannot bid more than μy. They therefore cannot offer the user cost required to outbid higher-income households for their assigned houses. Lower-income households either face the same local trade-off as in the unconstrained equilibrium or are constrained by an even lower borrowing limit, and therefore cannot profitably deviate upward either.

Finally, households with incomes below the participation threshold continue to choose the outside option. Hence the unconstrained assignment rule $q(y)$ remains an equilibrium assignment. □

We now prove Proposition 3.

Proof of Proposition 3. We first determine the tenure decision. A household with income y assigned to quality $q(y)$ is willing to pay at most

$$p^U(q(y)) - \tau$$

in rent to attain the same utility as in the unconstrained owner-occupied equilibrium. By the definition of y^τ, this rent exceeds the maximum user cost allowed by the borrowing constraint for $y < y^\tau$ and equals it at y^τ. Hence households with incomes below y^τ cannot purchase the houses assigned to them as owner-occupiers, but can consume them as renters.

Competition among tenants implies that buy-to-let investors charge the highest rent households are willing to pay, while competition among investors drives expected profits to zero. Thus, in the rental segment,

$$r(q(y)) = p^U(q(y)) - \tau, \qquad y \in [y^P, y^\tau).$$

For households with income at or above y^τ, the rent that compensates them for the disutility of renting is no larger than the user cost allowed by the borrowing constraint. Since these households prefer owning to renting, they remain owner-occupiers.

The owner-occupied user cost schedule starts from the boundary condition $p(q(y^\tau)) = \mu y^\tau$. If y'' exists, the schedule follows the borrowing constraint up to y'':

$$p(q(y)) = \mu y, \qquad y \in [y^\tau, y''].$$

For $y > y''$, the borrowing constraint is slack. The user cost schedule supporting the assignment $q(y)$ is then, similarly to Lemma 2:

$$\frac{dp(q(y))}{dy} = M(q(y), y - p(q(y))) \frac{f(y)}{g(q(y))}$$

with initial condition

$$p(q(y'')) = \mu y''.$$

Assumption 1 ensures that this schedule does not intersect the borrowing constraint again. If y'' does not exist, the borrowing constraint remains binding for all $y \geq y^\tau$.

It remains to verify that the assignment of households to housing qualities remains $q(y)$. Tenants face the rent schedule $p^U(q(y)) - \tau$, which differs from the unconstrained user cost schedule only by the constant τ. Hence their ranking over housing qualities is the same as in the unconstrained owner-occupied equilibrium, and the single-crossing property implies that no tenant can profitably deviate to another rental house.

Among owner-occupiers, the user cost schedule is constructed along the assignment $q(y)$. On intervals where the borrowing constraint is slack, households

are locally optimized at their assigned housing qualities; by the single-crossing property, this local optimality implies that no owner-occupier can profitably deviate to another housing quality. On intervals where the borrowing constraint binds, households would like to consume higher-quality housing, but cannot bid more than μy. They therefore cannot offer the user cost required to outbid higher-income households for their assigned houses.

Finally, households with incomes below the participation threshold continue to choose the outside option. Hence the assignment rule $q(y)$ is supported in equilibrium. □

A.2 Intertemporal utility maximization

The model discussed in the main text refers to a single period, but in this section of the Appendix we show that, under suitable assumptions, the static model can be interpreted as the intratemporal problem arising in a life-cycle optimization framework. We first present preferences, endowments and decisions that are common across environments. Then we show how the static model results from the environments without and with borrowing constraints.

A.2.1 Preferences, endowments and decisions

We consider an economy that consists of overlapping generations of households that live a known, finite number of periods $1, ..., T$. Households have preferences that are additively separable across periods and that can be represented by the utility function

$$U = \Sigma_{t=1}^{T} \beta^{t-1} u(q, c), \tag{A3}$$

where $u(q, c)$ is the same periodical utility function as in the static model. $\beta \in (0, 1)$ denotes the discount factor. Households have no bequest motive.

Households are endowed with a deterministic stream of earnings $\{e_1, ..., e_t, ..., e_T\}$ that is non-decreasing across periods: $e_t \geq e_{t-1}$. Households start their life without any wealth. When a household dies, any wealth remaining after repayment of debts is lost. Deceased households are replaced by newborns, so that the environment is stationary. In particular, the aggregate earnings distribution is constant, just as the house quality distribution.

Each period, households choose their house quality and other consumption. The consumption of owner-occupied housing requires buying a house, and changing housing consumption across periods requires both buying and selling. However, because there are no transaction costs, households can re-optimize every period. Moreover, because the earnings and house quality distributions are constant, house prices are constant over time.

Households have the possibility to save, but can only borrow for the purchase of a house. To finance the purchase of a house, the market offers interest-only

mortgage loans with a loan-to-value ratio of 1. The mortgage interest rate is denoted by i and for simplicity there are no other costs associated with housing. The mortgage interest rate i equals the interest rate on savings and reflects households' rate of time preference, $\beta = 1/(1+i)$.

A.2.2 No borrowing constraints

Consider the environment of Section 2.2 in which households have access to mortgages with an unconstrained amount M, as long as M equals the price $P(q)$ of the house that they buy. Of course, when households sell their house, they also need to repay their mortgage. Denoting accumulated savings at the beginning of period t by b_{t-1}, the budget constraint of a household in period t thus equals

$$\begin{aligned} c_t + P(q_t) - M_t + iM_t + b_t &\leq e_t + b_{t-1} + P(q_{t-1}) - M_{t-1}, \\ c_t + p(q_t) + b_t &\leq e_t + b_{t-1}, \end{aligned} \tag{A4}$$

because $P(q_t) = M_t$ and $iP(q_t) = p(q_t)$, and where $b_t \geq 0$ and $b_0 = 0$. Because we ignore other costs associated with housing, housing expenditure is thus equal to the amount of interest paid on a loan that equals the price of the house. Effectively, durable housing thus turns into a nondurable good.

Because $\beta = 1/(1+i)$, each period households would like to consume the permanent component of their lifetime earnings. However, because their stream of earnings is non-decreasing and households cannot borrow for other consumption, $b_t = 0$ in every period. The budget in each period thus simply consists of earnings e_t, and we can think of the income y in the static model as e_t. If earnings are increasing over the life-cycle, consumption of housing and other consumption are thus increasing as well.

Because $\beta = 1/(1+i)$, households would optimally smooth marginal utility over time. Since earnings are non-decreasing and households cannot borrow against future labor income except through mortgages, the non-negativity constraint $b_t \geq 0$ binds whenever desired consumption exceeds current earnings. Under these assumptions, households optimally choose $b_t = 0$ in every period, so current expenditure is financed entirely from current earnings. Consequently, the relevant budget constraint in each period reduces to

$$c_t + p(q_t) \leq e_t,$$

which is identical to the budget constraint in the static model with $y = e_t$.

The intratemporal first-order conditions of the maximization of (A3) subject to (A4) equal the first-order condition of the static problem in (10) for every period. We can thus conclude that the static model in the body of the paper can be embedded in a life-cycle utility maximization framework.[25]

[25]Some of the assumptions above can be relaxed. For instance, the conclusion remains valid if the

A.2.3 Uniform borrowing constraints

Now consider an environment with a maximum mortgage payment-to-income constraint. Households can still save and borrow for housing, but now the maximum mortgage loan available to a household in period t is a function of its earnings e_t. In particular, households have access to interest-only mortgage loans with interest rate i up to size $\mu e_t / i$, and do not have access to a larger mortgage. Consequently, they cannot finance the purchase of a house with price exceeding $\mu e_t / i$ by relying on current income. This reflects the interpretation of payment-to-income constraints as limits on the amount of debt that can be serviced out of current earnings. While households may use accumulated wealth to reduce the required mortgage, they cannot obtain a larger mortgage by committing a greater share of current income, since that same income must be available to service the loan.

The relevant case is the one in which the mortgage constraint binds. A household could then depart from the behavior assumed in the main text in two ways. First, it could purchase a more expensive house without a mortgage by paying the additional purchase price out of current income. However, this increases the marginal cost of housing from its user cost $p(q) = iP(q)$ to its purchase price $P(q)$. Since $i < 1$, the trade-off between housing and other consumption becomes substantially steeper. For realistic interest rates, purchasing housing beyond the mortgage limit out of current income is therefore unattractive unless households place an implausibly high marginal value on additional housing quality.

Second, a constrained household could in principle save over several periods and later use accumulated wealth to purchase a house whose price exceeds the mortgage limit. We abstract from this dynamic saving motive. A sufficient condition for this deviation not to be optimal is that the utility cost of the required saving exceeds the discounted value of the future housing gain. Let q^C denote the best house that can be financed immediately under the borrowing constraint, and let $q^H > q^C$ denote a house that could only be purchased after accumulating wealth. If saving toward q^H requires the household to choose a lower current allocation (q^S, c_t^S), then the deviation is not profitable whenever

$$u(q^C, c_t^C) - u(q^S, c_t^S) \geq \frac{\beta(1-\beta^{T-t})}{1-\beta} \Delta u^{\max}(q^H, q^C),$$

where $\Delta u^{\max}(q^H, q^C)$ is an upper bound on the per-period utility gain from consuming q^H rather than q^C. The finite horizon limits the value of circumventing the borrowing constraint: the closer the household is to the end of life, the fewer periods remain over which the higher housing quality can be enjoyed. Under this condition, constrained households optimally accept the mortgage limit rather than saving in order to relax it later.

rate of time preference is higher than the interest rate or if the periodical utility functions change over time.

A.3 Additional case uniform borrowing constraint

This section illustrates a limiting case of Corollary 1. In this case, the constrained user cost schedule coincides with the borrowing constraint only at the participation threshold y^P and lies strictly below it for all higher income levels, as illustrated by the gray curve in Figure A1.

This case arises when

$$M(q(y), (1-\mu)y)\frac{f(y)}{g(q(y))} < \mu$$

holds immediately to the right of y^P. Equivalently, households' marginal willingness to pay for housing falls below the marginal increase in borrowing capacity as soon as the borrowing constraint becomes binding. Consequently, the constrained user cost schedule coincides with the borrowing constraint only at y^P and immediately departs from it thereafter.

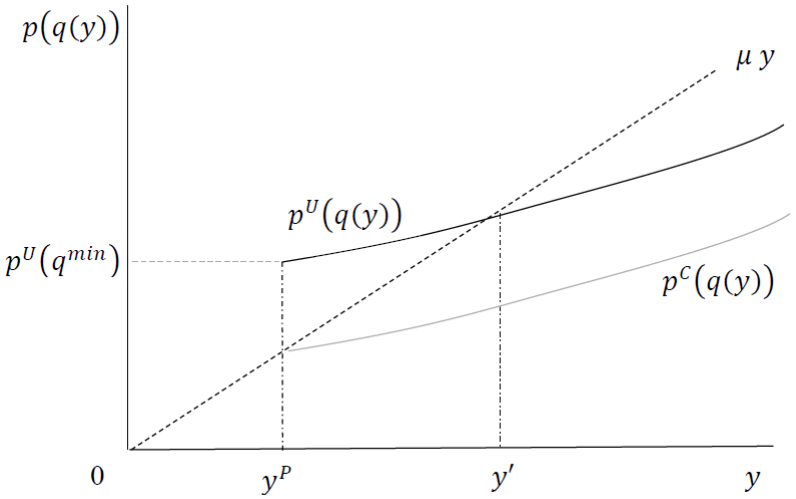


Figure A1: The mortgage payment-to-income constraint and the user cost of housing with a small marginal willingness to pay for housing.

A.4 Extension: heterogeneous borrowing constraints

In this subsection we consider a more general situation in which households can experience borrowing constraints of a general nature. The borrowing constraint is a household-specific maximum imposed on the purchase price of a house.[26] Since there is a one-to-one relationship between purchase price and user cost, we include this in our model as a maximum user cost ρ. We assume that ρ has positive support on $[\rho^{\min}, \rho^{\max}]$ for some $0 \leq \rho^{\min} < \rho^{\max} \leq \infty$.

[26] This household-specific maximum can be conditional on the mortgage underwriting rules and the mortgage interest rate.

Households are characterized by their income and maximum user cost, an ordered pair (y, ρ). We denote the simultaneous distribution of income and maximum user cost as $F(y, \rho)$. $F(y, \rho)$ is thus the number of households with income at most equal to y who can bid at most ρ for a house. The corresponding density is $f(y, \rho)$. In the previous subsection we discussed a case in which ρ is a function of income, $\rho = \mu y$. This is a special case of the situation considered here, in which the density $f(y, \rho)$ is only positive for $\rho = \mu y$. The discussion that follows considers a different and much more general case in which $f(y, \rho)$ is a continuous function of its two arguments. Such a situation is compatible with the mortgage constraint discussed in the previous section if households also have wealth that can be used, in addition to a mortgage loan to finance a house. The wealth distribution is continuous and may be income-specific.

The ρ that is relevant for a particular household should be interpreted as the maximum user cost that the household can afford. For instance, if the mortgage payment-to-income constraint of the previous section holds, this allows a user cost $\rho^m(y)) = \mu y$, If the household also has wealth that can be used to help finance the house, there is a second part of the user cost, denoted ρ^w which is a function of the household's wealth w.[27] The constraint is now that the actual user cost p is at most equal to the sum $\rho = \rho^m + \rho^w$.

Note that the analysis is also relevant for other cases. For instance, if there is a down-payment constraint (implied by a maximum LTV-ratio) instead of a mortgage payment-to-income constraint, the household must have enough wealth to pay a share σ that equals one minus the maximum loan-to-value ratio. Using the assumptions of the previous paragraph, the maximum user cost may be determined as follows. The down-payment constraint is: $\sigma P < w$, where P denotes the purchase price. Multiplication of both sides of the inequality by the mortgage interest rate and dividing by σ gives: $rP < rw/\sigma$. The left-hand side of this inequality is the user cost, while the right-hand side gives the maximum of the user cost, ρ, as implied by the downpayment constraint. It does not depend on income, only on wealth. The model discussed here is therefore also relevant for situations with a down-payment constraint.

The supply side of the market is unchanged. The number of households assigned to a house must be equal to the number of houses that is available. These households must have an income that is at least as high as the critical level at which housing of the lowest quality is consumed and a maximum user cost that is larger than that of the lowest quality housing. There are thus potentially two groups of households demanding housing of minimum quality: (i) those with a maximum

[27] For concreteness, one may assume that user cost is proportional to the purchase price of a house, with the constant of proportionality equal to the mortgage interest rate for mortgage financing. If it is further assumed that the opportunity cost of wealth is equal to this interest rate there is a conveniently simple relationship between the purchase price of a house and its user cost. The discussion below covers to this case.

user cost $\rho = p(q^{\min})$ and income $y \geq y^P$, and (ii) those with income $y = y^P$ and maximum user cost $\rho \geq p(q^{\min})$:

$$F(y^{\max}, \rho^{\max}) - F(y^P, \rho^{\max}) - F(y^{\max}, p(q^{\min})) + F(y^P, p(q^{\min})) = S, \tag{A5}$$

in which the last term shows up to avoid double-counting.

The value of $p(q^{\min})$ is determined in the same way as before, namely by the condition in (8) that a household with the critical income must be indifferent between the housing of minimum quality and the outside option. Because housing is a normal good, $p(q^{\min})$ is an increasing function of the critical income y^P, and the number of households with a maximum user cost above $p(q^{\min})$ must thus be a decreasing function of y^P. It follows that the left-hand side of (A5) is a decreasing function of y^P and that y^P is uniquely determined.

To trace out the user cost function we consider what happens at a combination of income y, quality q and an associated user cost $p(q)$. The idea is that all households with an income lower than y or a maximum user cost lower than $p(q)$ either have been assigned a house, or will not participate in the housing market. The supply of housing of quality q is $g(q)dq$ and this must be equal to the demand. Demand originates both from households experiencing borrowing constraints and from those who do not, so that

$$g(q)dq = f^{bc}(y, p(q))dp + f^{uc}(y, p(q))dy, \tag{A6}$$

in which $f^{bc}(y, p(q))$ is the density of households who have not been assigned a house but are constrained at user cost $p(q)$,

$$f^{bc}(y, p(q)) = \int_y^{y^{\max}} f(y, p(q))dy, \tag{A7}$$

while $f^{uc}(y, p(q))$ is the density of unconstrained households choosing a house with quality q,

$$f^{uc}(y, p(q)) = \int_{p(q)}^{\rho^{\max}} f(y, \rho)d\rho. \tag{A8}$$

Figure A2 illustrates. The box indicates the combinations of income $y \in [y^{\min}, y^{\max}]$ and maximum user costs $\rho \in [\rho^{\min}, \rho^{\max}]$ for which the distribution $F(y, \rho)$ has positive support. The housing price is given as a function of income. It starts at the critical income and is shown until some higher y corresponding to housing demand q that commands price $p(q)$. The two narrow (blue) rectangles indicate the demand for housing at this point. The vertical one refers to unconstrained households, that is households with income y who are able to bid at least $p(q)$. The horizontal box refers to constrained households, who can just afford to bid $p(q)$ but cannot afford more expensive housing because of a borrowing constraint. Total demand for housing of quality q is equal to the number of households whose combinations of income and maximum loan belong to these two boxes.

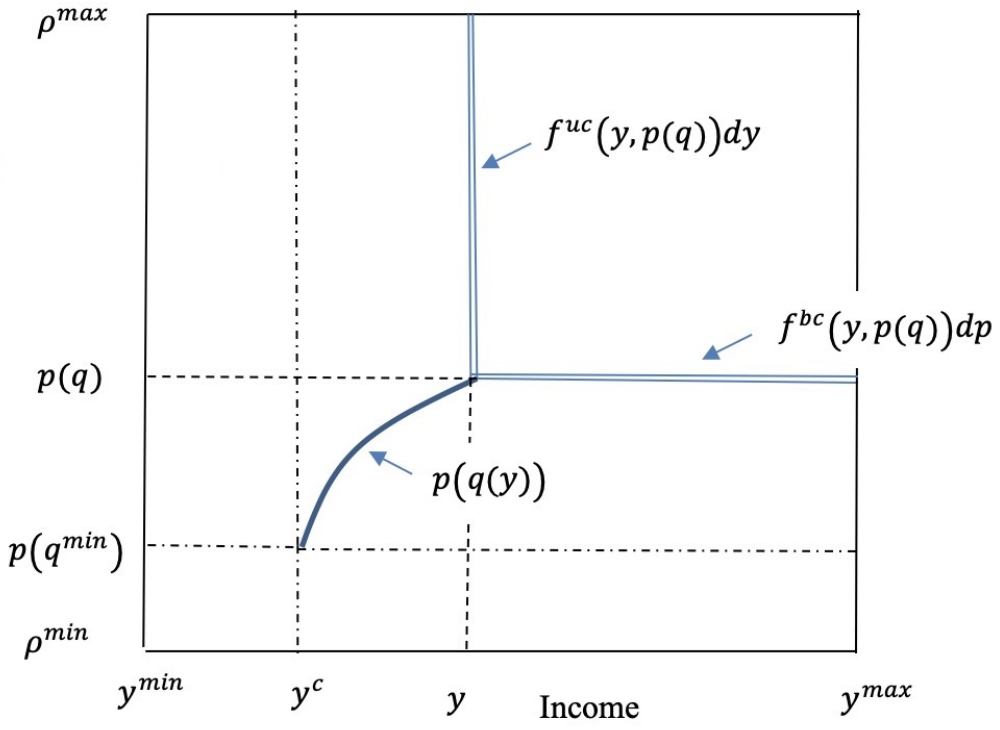


Figure A2: Constrained and unconstrained housing demand

Observe that unconstrained households will only choose the combination $(q, p(q))$ of housing quality and user cost if the first-order condition (10) holds. Using the definition of the marginal price $\pi(q) = dp/dq$, participation of unconstrained households at $(q, p(q))$ thus requires

$$dp = M(q, y - p(q))dq. \tag{A9}$$

Substituting (A9), we can rewrite (A6) as

$$[g(q) - f^{bc}(y, p(q))M(q, y - p(q))]dq = f^{uc}(y, p(q))dy. \tag{A10}$$

Since the right-hand side is non-negative, the expression in square brackets on the left-hand side must also be non-negative. If this is the case, there is a 'mixed' local equilibrium in which a given type of housing is inhabited by both types of households, a situation that did not occur with the uniform mortgage payment-to-income constraint studied in the previous subsection.

However, if the expression in square brackets on the left-hand side of (A10) is negative, (A9) does not hold at $(q, p(q))$ and a mixed equilibrium is not feasible. In this situation, there are so many households with a binding borrowing constraint at $p(q)$ that they consume all housing with quality q and nothing is left for unconstrained households. The second term on the right-hand side of (A6) thus disappears ($f^{uc} = 0$), and instead we have

$$g(q)dq = f^{bc}(y, p(q))dp. \tag{A11}$$

The slope of the user cost function will now be determined by the densities of housing and of borrowing-constrained households in such a way that all constrained households are exactly on their constraint:

$$\pi(q)(=\frac{dp}{dq}) = \frac{g(q)}{f^{bc}(y, p(q))}. \tag{A12}$$

Since all households are constrained, the marginal price of housing is lower than the marginal willingness to pay.[28]

Note that (A11) allows the density of borrowing-constrained households to be larger than the density of housing. If many borrowing-constrained households are clustered in a particular price-quality range, they may occupy all housing for a range of qualities, as happened in the previous subsection. Unconstrained households will be put 'on hold' until all the constrained households have been served. This implies that these unconstrained households are directed to higher quality housing. Because constrained households are forced to accept lower housing consumption, unconstrained households are enabled to consume housing of higher quality relative to the unconstrained equilibrium.

As will be clear by now, the equilibrium with borrowing constraints in the model of the present subsection differs substantially from that in the previous subsection with a uniform mortgage payment-to-income constraint. In particular, the assignment rule in the constrained equilibrium now differs from that in the unconstrained one. Households experiencing a binding borrowing constraint will in general consume less housing than unconstrained households with the same income level. Some of them may even be pushed out of the housing market, while other households with lower incomes but less tight borrowing constraints will be able to enter. However, with the generalized borrowing constraints of the present subsection it is still true that - relative to the corresponding equilibrium without constraints, house prices will be lower.

Lemma 5. *In the equilibrium with general borrowing constraints (i) housing consumption of an unconstrained household consuming quality q^* in the corresponding unconstrained equilibrium will never be lower in the constrained equilibrium and strictly higher if households with a higher income have been allocated to housing quality that is lower than q^* , (ii) house prices $p(q)$ will never be higher than in the corresponding unconstrained equilibrium and strictly lower if some households consumer lower housing quality than $q*$ are constrained.*

$$-\frac{d\pi}{dq} < -\frac{dM}{dq}\bigg|_{u\ constant} \tag{A13}$$

Proof. (i) Since all houses are occupied relegating constrained households to lower quality means that unconstrained ones must move to higher quality. Note that for

[28] To verify this, note that the expression in square brackets on the left-hand side of (A10) is negative implying that $\frac{g(q)}{f^{bc}(y,p(q))} < M(q, y - p(q))$ and use (A12).

unconstrained households a higher income still implies higher housing consumption. (ii) Suppose the price function would remain unchanged. Then all unconstrained households that have moved to higher quality have lower incomes than the households living in the same houses in the situation without borrowing constraints. Their marginal willingness to pay for housing is lower. The marginal price for houses only occupied by constrained households is also lower than that on the unconstrained households that were put on hold for that quality. Constrained households thus imply lower housing prices, also for the households that they push to higher quality. That leaves us with the possibility that for higher qualities some households live in the same houses as in the situation without constraints. The housing price of the house of the lowest quality occupied by these households is lower than in the situation without borrowing constraints. The marginal willingness to pay for housing is therefore higher, which moves the housing price towards the one in the situation without borrowing constraints for higher qualities, but it will never exceed this price. The situation is similar to that with a uniform borrowing constraint. □

For our empirical work an important implication of the generalization of the assignment model to situations in which households have wealth and can use it to avoid borrowing constraints that would otherwise be binding is that ignoring wealth when assigning households to the treatment group leads to overestimation of the size of this group and therefore overestimation of the possibilities for arbitrage by buy-to-let investors. An important new element is that the presence of binding borrowing constraints will now lead to changes in the assignment rule, which is no longer purely based on income. Since some households are able to avoid or mitigate the binding borrowing constraint by using their wealth to finance housing, the price decrease associated with a given constraint is lower than in the situation in which no household has wealth, which further limits the scope for buy-to-let arbitrage. Note that in reality households do not only use their own wealth, but sometimes also have access to that of parents or other relatives to help finance their house when constraints are binding.

A.5 Extension: Taste heterogeneity

A.5.1 Introduction

The assumption that all households have the same tastes appears unrealistic and we will therefore in this section consider what changes in the model if we allow heterogeneity in tastes. More specifically, we assume that there are $n \geq 2$ groups of households. Within each group all households have the same tastes, but their incomes differ. We use a super-fix $i = 1...n$ to refer to groups. Incomes belong to group-specific intervals $[y^{(i,min)}, y^{(i,max)}]$ and the income distributions $F^i(y^i)$ are differentiable and increasing on that interval. The total number of households in

all groups exceeds the number of houses. Moreover, the housing distribution $G(q)$ is differentiable and increasing on that same interval. For each group the same regularity assumptions hold as were assumed in the main text for the situation with a single group. Hence there would exist a market equilibrium for each of the groups separately if the housing stock were smaller than the number of households in this group. We refer to the situation just defined as the assignment model with multiple groups.

To provide some intuition for the results that follow, it is helpful to recall the familiar graphical analysis associated with Rosen (1974) in which households that differ in tastes or incomes maximize utility while taking a hedonic price function, referring to a differentiated good with a single characteristic, as given. Different households in general choose different positions, but it is possible that households with different tastes and incomes choose the same position on the hedonic price function. The necessary condition for optimal choice is that the marginal willingness to pay for the characteristic equals its marginal price, while the sufficient condition is that the household's indifference curve is more convex than the budget constraint. Market equilibrium requires that all varieties present in the market are chosen by some households and that for each variety there be an equilibrium between supply and demand.

In the subsections that follow we discuss various aspects of the assignment model with taste heterogeneity. We start with considering the situation in which members of two or more groups demand the same housing qualities and show how we can still trace out the housing price function. The analysis reveals that with multiple groups the second order condition may be violated for one or more (but not all) groups for specific housing qualities. If this happens, members of such a group do not express demand until the condition is satisfied again. In the subsection that follows we show that in general there is only one group that demands the houses of the lowest quality, while the others will subsequently join.

A.5.2 When multiple groups demand the same houses

In this subsection we describe how we still trace out the housing price function when households from two or more groups may express demand for the same houses. That is, for a given housing quality and at a given price, households belonging to two or more groups have the same marginal willingness to pay for housing. The assignment of households to housing is now described by:

$$g(q)dq = \sum_i \delta^i(q) f^i(y) dy^i \tag{A14}$$

In this equation $\delta^i(q)$ is a 0-1 variable indicating that households of type i chose houses with quality q. For $\delta^i(q)$ to be equal to 1, the following conditions have to be satisfied (i) the house offers a utility that exceeds that of the outside option,

(ii) the first-order condition is satisfied for an income $y \in [y^{i,min}, y^{i,max}]$ and (iii) the second-order condition is also satisfied. The equation states that all available houses of quality q will be occupied by households that reach their optimal housing demand there.

This equation does not yet make clear how the houses are distributed over the households of the various groups. To address this issue we consider the change in the first-order condition (10) that occurs if we move to a slightly higher housing quality. If households of group i continue to express demand at this higher quality, the equality between marginal willingness to pay for housing and the marginal price of housing must be maintained. This requires: $d\pi/dq = dM^i/dq$ for all groups that continue to express demand. Elaboration of this condition gives:

$$\frac{d\pi}{dq} = [\frac{\partial M^i}{\partial q} - \pi \frac{\partial M^i}{\partial (y-p)}] + \frac{\partial M^i}{\partial (y-p)} \frac{dy^i}{dq} \tag{A15}$$

The expression in square brackets on the right-hand side is closely related to the second-order condition of 4, which requires:

$$\frac{d\pi}{dq} \geq [\frac{\partial M^i}{\partial q} - \pi \frac{\partial M^i}{\partial (y-p)}] \tag{A16}$$

If the first-order condition is satisfied, but the second-order condition fails, households express no demand. If the first as well as the second order condition is satisfied, equation (A15) implies that $dy^i/dq > 0$. Since the change in the marginal price of housing must be the same for all households the changes in income associated with the movement to a higher housing quality for the different groups must be aligned to each other. To see how this works, we solve (A15) for dy^i/dq and substitute the result into (A14). This gives us an expression for the change in the marginal price of housing that is compatible with equilibrium:

$$\frac{d\pi}{dq} = \frac{\sum_i \delta^i(q) f^i(y^i) (\frac{\partial M^i}{\partial (y^i-p)})^{-1} [\frac{\partial M^i}{\partial q} - \pi \frac{\partial M^i}{\partial (y^i-p)}] + g(q)}{\sum_i \delta^i(q) f^i(y^i) (\frac{\partial M^i}{\partial (y^i-p)})^{-1}} \tag{A17}$$

The change in the marginal price implied by (A17) can now be used to find the change in income for each group of households that expresses demand.

We can thus still trace out the demand function when households with different tastes express demand for housing of the same quality. Households of a particular group that express demand for housing of a particular quality will only stop doing so for higher qualities if the maximum income is reached or if the second-order condition is no longer satisfied. In such a case the households can always reach a higher utility by putting their demand on hold. Their utility will increase if the housing price function is traced out further.

A.5.3 Demand at the lowest housing quality

As in the case with homogeneous households, the price function starts at the housing of minimum quality. With multiple groups of households we can no longer

determine a priori what will be the income of the household occupying this type of housing, nor to which group or groups these households belong and then determine its price. We will instead, for the moment, take this price $p(q^{\min})$ as given, and discuss its determination in the next subsection.

There households that occupy the housing of minimum quality must be indifferent between the outside option, which we still assume to have a given price and quality, and the housing of minimum quality. Setting the utilities of the outside option and the housing of minimum quality equal to each other allows us to derive the critical income for each group. It may happen that for one or more groups this income is higher than the maximum or lower than the minimum. For an equilibrium, there must be at least one group for which the critical income is in the support of the relevant income distribution. If there is exactly one, the allocation procedure starts as in the case with homogeneous tastes. If there are two or more, we have to make a choice, because the marginal willingness to pay for housing at the housing of minimum quality will in general be different for members of different groups.

We select the one with the lowest value of the marginal willingness to pay M at the housing of minimum quality and start the allocation process with this group. That is, we set the marginal price equal to the marginal willingness to pay for the selected group and start tracing out the price function as if it were the only one. [29] The households with the critical income in other groups have a willingness to pay for housing of the minimum quality that exceeds the marginal price. Their utility will increase if they put their demand on hold and follow the tracing out of the housing price function until it becomes tangent to their bid price function. [30]

At the minimum housing quality there may be households of other groups that are not indifferent between this housing and the outside option, but that have a marginal willingness to pay for housing that equals the marginal price. These households must have a lower income than the critical level of their group and will therefore prefer the outside option to the housing of minimum quality.

However, when the allocation procedure proceeds and higher levels of housing quality are reached, it may happen that the marginal price of housing is equal to the marginal willingness to pay for housing of a household belonging to a group that is thus far not involved in the allocation process, that is indifferent between this type of housing and the outside option. As soon as this happens, this group will start participating in the allocation process and the procedure outlined in the

[29] If there are two or more groups with the lowest marginal willingness to pay, start tracing out the price function with all of them simultaneously as discussed above.

[30] Note that if we would start tracing out the price function with a group that had a higher willingness to pay for the housing of minimum quality, the group with the lowest marginal willingness to pay would be excluded from housing. However, this is incompatible with equilibrium, because there are households of this group with higher incomes who strictly prefer housing of minimum quality to the outside alternative. They will drive up the price of the housing of minimum quality.

previous subsection now becomes the relevant one. Other groups may join at other levels of housing quality.

Note that the incomes at which these groups start expressing demand for housing will in general be lower than the critical income as it was defined above, where it referred to housing of minimum quality. Since housing quality is not higher, households may be indifferent between this housing and the outside option at the prevailing price at a lower income.

By continuing the allocation procedure in this way, we may trace out the price function for all housing qualities. There may be some intervals at which one or more groups express no demand because the second-order condition is not satisfied and ultimately for all groups the households with the highest incomes are allocated. If the house price function corresponds to a market equilibrium the households with the highest incomes of the remaining groups are allocated to the houses of the highest quality. [31]

A.5.4 Existence of a market equilibrium

In the previous subsections we have described an equilibrium with multiple groups. In the present subsection we show that such an equilibrium exists. We start with two lemmas in hwich we consider the price of the housing of minimum quality $p(q^{\min})$ as a choice variable denoted as p^s. By starting the allocation procedure from an arbitrary value of p^s, it will end either before all households are allocated or before all houses are allocated. However, we will show that a market equilibrium can be reached by appropriately choosing its value.

Lemma 6. *Lemma. If we start the allocation process from a higher value of p^s, the price housing will be higher for all qualities. Moreover, the price of housing of any quality is a continuous function of the starting price p^s.*

Proof. Suppose that the first statement is false. Then there must be a housing quality for which the price is lower. The two price functions must have crossed at some housing quality, say $q^@$. There the slope of the price function starting from the higher price must have been lower than that of the price function starting from the lower price. That slope is equal to the marginal willingness to pay of the households allocated to that quality. We know that for at least one group the income of the households allocated to quality $q^@$ must be higher if the price function starting from the higher p^s is relevant. However, for the same price $p(q^@)$ the marginal

[31] It is possible that all groups involved in the allocation process (that all groups for which $delta^i(q) = 1$) stop expressing demand because their maximum income is reached. Demand for the housing that is still vacant can then only result from households that were not involved in the last phase of the allocation procedure as realized thus far. The allocation procedure should then be re-started by selecting the group with the lowest marginal willingness to pay for housing at the price-quality combination at which the process stopped, and continued in the same way. In this exceptional situation their will be a kink in the price function.

willingness to pay for housing must be higher for the higher income households. This implies a contradiction with the flatter slope of the price function starting from the higher p^s. Suppose that the second statement is false. Then there must be at least one discontinuity in the price function: for a certain quality, say $q^@$ and a starting price $p^{s@}$ the price function jumps upwards by a strictly positive amount, no matter how small the increase in the starting price is. However, this implies an immediate contradiction with the fact that the increase $dp(q^@)$ is equal to the marginal willingness to pay for housing, which is a continuous function of the income of the households occupying the houses of that quality, while this income is a continuous function of p^s. □

Now define the thresold income level $y^{(i,t)}(p^s)$ of group $i, i = 1...n$ as the lowest income for which this group expresses demand for housing. All households with income higher than the threshold level will consume housing. Existence of a market equilibrium thus requires that the sum of all households with at least the critical income is equal to the number of houses.

Lemma 7. *Lemma. The threshold incomes $y^{(i,c)}(p^s)$ are continuous increasing functions of the price p^s of minimum quality housing.*

Proof. At the critical income, a household is indifferent between the outside option and the preferred housing quality. For a higher price of that housing the household prefers the outside option. All lower incomes already did so, and will not change. Hence the critical income must increase. Continuity of the threshold income in the starting price p^s is obvious for the groups that consume housing of minimum quality. So consider a group for which the critical income chooses a housing quality larger than $q^{\min}$, say $q^@$. Because of the higher price $p(q^{\min})$ the housing price function shifts upwards. The initial critical income will no longer result in indifference between housing of quality $q^@$ and the outside alternative. However, a higher income will do this because the price function is continuous. The indifference curve corresponding to this higher income may not be tangent to the new price function, at the same quality. However, the quality at which the tangency will hold must be close to $q^@$ and approach that value if the change in the starting price gets arbitrarily small. □

To establish existence of a market equilibrium, we must demonstrate the existence of a price p^s for housing of minimum quality that implies values of the critical incomes of all groups for which the total number of households with incomes at least equal to the critical income of their group is equal to the housing stock.

Assumption A1. If the price of the outside option and the price of minimum quality housing are equal, the sum over all groups of the number of households preferring housing of minimum quality exceeds the housing stock.

Assumption A2. If the price of housing of minimum quality p^s is sufficiently high, all households prefer the outside option.

Proposition 4. *If Assumptions A1 and A2 hold, there exist a unique market equilibrium in the assignment model with multiple groups.*

Proof. The critical incomes are continuously increasing functions of the price p^s that are equal to $y^{i,min}$ when $p^s = p^*$ and to $y^{i,max}$ if that price is sufficiently high. Let y^h be the total number of households (in all groups) with income at least equal to the critical value of their group. It follows that y^h is a continuously decreasing function of p^s that is equal to the total number of households in all groups for $p^s = p^*$ and 0 for p^s sufficiently high. Somewhere between these two prices must be a value $p^s = p^{eq}$ for which y^h equals the total number of houses. □

A.5.5 A special case

It may be noted that two different aspects of heterogeneity are important in the analysis offered above. One is that households of different groups may realize the same marginal utility of housing at different income levels, the other is that the convexity of the indifference curves (the first derivative of the marginal willingness to pay for housing) may be different even though households belonging to different groups have the same marginal willingness to pay for housing. It is possible that the first aspect is relevant, while the second is not. That situation occurs if the indifference curves of households belonging to different groups are parallel to each other, that is if $u^i(c,q) = u(\theta^i + c, q)$. Two households belonging to different groups choose exactly the same location on the hedonic price function if the difference between their incomes is equal to minus the difference between their θ s.[32]

When a particular value of housing consumption q is chosen by households of group i with income y, that same value of housing consumption will also be chosen by households of group j with income $y - \theta^i + \theta^j$. Note that the share of households of the two groups may still vary over housing qualities, depending on their income distributions.

To illustrate consider a population where all households have a linear demand function for housing, with a group-specific intercept:

$$q = \alpha^i + \beta\pi + \gamma y, \tag{A18}$$

differences in the constant term α^i have a similar impact on the demand for housing as differences in income. Hence in equilibrium households with different income levels will occupy the same houses, as we see in reality. The demand function refers to a linear budget constraint, whereas in the assignment model the budget constraint may be nonlinear. However, with a nonlinear budget constraint it is still

[32]To see this, consider two groups of households $i = 1,$. Their budget constraints are $y^i = c + p(q)$. Substitution in the utility functions gives: $u^1 = u(\theta^1 + y^1 - p(q))$ and $u^2 = u(\theta^2 + y^2 - p(q))$. The utility functions are therefore identical when $y^1 - y^2 = -(\theta^1 - \theta^2)$.

the case that housing consumption is always identical for households belonging to different groups, say 1 and 2, if $y^1 - y^2 = -(\alpha^1 - \alpha^2)/\gamma$.[33]

Other consumption is equal for both households if $\theta^1 - \theta^2 = (\alpha^1 - \alpha^2)/\gamma$. Although the direct utility function that generates the linear demand curve is unknown, we can therefore be sure that we are in the situation described above, where a group-specific constant is added to other consumption in a utility function that is otherwise identical for all groups.

A.5.6 Borrowing constraints

What happens if we have multiple groups of households and borrowing constraints? Even if the borrowing constraints are homogeneous in the sense that they are always given by the same function of income, the taste differences imply that their impact will differ over the groups. If houses of all qualities are occupied by households from different groups and some of them are constrained and others not, the constrained households have to reduce their housing consumption, which implies that the unconstrained households may shift to higher quality levels. This is qualitatively similar to what happened when we introduced wealth into the model. Indeed, Lemma 5 continues to hold in the situation with heterogeneous tastes.

B Data construction

We use transactions of existing, initially owner-occupied housing as provided by the Land Registry (Kadaster).We obtain changes in ownership status of those properties in subsequent years through Eigendomtab. We can infer the addresses and individuals to which these transactions correspond through GBAAdresobjectbus, and link these to households using GBAPersoontab and GBAHuishoudensbus. We match households' total mortgage loans to each buyer and house using the information on household assets in Vehtab. Via Inhatab and Inpatab, we obtain the gross income of the household and its individual members involved in the transaction.

There were 411k transactions of owner-occupied housing in 2009 and 2010. Out of these, we can successfully match 283k transactions to buyer income information, and 122k transactions to buyer loan information. We retained only those buyers with a first-time home mortgage, identifying them as first-time buyers. Subsequently, by combining buyers' loan and income information, we calculated the

[33] To see this, note that with a nonlinear budget constraint $y = c + p(q)$ demand can be written as $q = \alpha^1 + \beta\pi + \gamma(y - p(q) + piq)$ where the expression in brackets denotes the 'virtual income.' This equation is based on a linearization of the nonlinear budget constraint at the optimal consumption of housing and other goods. It is easy to verify that for two groups $i = 1, 2$ housing consumption is still equal if $y^1 - y^2 = -(\alpha^1 - \alpha^2)/\gamma$.

buyers' MPTI, resulting in approximately 64k valid transactions. We removed transactions that had already violated the 2011 NIBUD regulations, defining them as the "always treated" sample. This process resulted in our final baseline sample of approximately 51k transactions. Table A1 presents the summary statistics of our baseline sample. Figure A3 shows the distribution of the corresponding MPTI ratios.

We then transform this housing sample into a panel dataset covering the period from 2011 to 2017. We exclude over 2k observations that dropped out of the treatment group in 2017. We also removed over 1k observations for which the ownership status was unknown in that year. We performed dual validation on the property ownership data, ensuring that the registered occupant of the house in a given year is consistent with the property ownership. As a result, we excluded a small number of inconsistent cases. We finally arrive at 346,489 observations, which is slightly lower than the number of observations in a balanced panel dataset (50,657*7=354,599). Table A2 gives the summary statistics of the ownership status and violation status of the panel.

C Additional empirical results

C.1 Additional tables

Table A1: Summary Statistics of Baseline Transactions

Variable	N	Mean	SD	p5	p95
MPTI	50,657	0.220	0.056	0.122	0.305
Sale Price	50,657	207,367	92,545	112,000	357,000
Loan Amount	50,657	213,477	92,774	105,998	356,762
Household Income	50,657	66,204	33,052	32,913	116,598
Household Wealth	50,657	32,465	174,625	-12,579	121,823
Property Type	50,657	0.297	0.457	0	1

Notes: The table reports summary statistics for the baseline transactions. Mortgage payments used to construct MPTI are calculated according to equation (16). Household income refers to annual gross household income; loan amount refers to the total loan amount at the end of the year of origination; and household wealth refers to wealth excluding owner-occupied housing. Property Type equals one for apartments and zero for single-family houses.

Table A2: Summary Statistics of the Panel Dataset

Variable	N	Mean	SD
Private Rent	346,489	3.320	17.915
Private Rent with Transaction	346,489	1.092	10.393
Private Rent without Transaction	346,489	2.227	14.758
Violation	346,489	0.057	0.232
MPTI	346,489	0.207	0.054

Notes: The table reports summary statistics for the property-year panel. Private Rent equals 100 if a property is privately rented in a given year and zero otherwise. Private Rent with Transaction equals 100 if a property is privately rented and transacted in the same year, while Private Rent without Transaction equals 100 if a property is privately rented but not transacted in that year. MPTI is the ratio of annual mortgage payments to annual household income. Following Van Bekkum et al. (2024), we adjust loan amounts to account for the tightening of maximum loan-to-value (LTV) limits over the sample period. Starting from a median LTV of 110%, we reduce the LTV to 106% in 2011 and 2012 and subsequently by one percentage point per year from 2013 to 2017, reaching 101% in 2017. These adjustments are consistent with changes in the median LTV observed in the data and the evidence in Van Bekkum et al. (2024). As a result, the average MPTI decreases from 0.220 in the baseline transactions (Table A1) to 0.207 in the property-year panel.

Table A3: Alternative Estimators of Baseline Difference-in-Differences Estimates

	Private Rental							
	SA				BJS			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Violation	1.369***	1.048***	1.072***	0.921***	1.294***	1.017***	1.003***	0.888***
	(0.254)	(0.263)	(0.251)	(0.262)	(0.254)	(0.263)	(0.252)	(0.263)
Type × Year trend			1.653***	1.387***				
			(0.052)	(0.071)				
N	346,470	343,336	346,470	343,336	346,427	346,266	346,427	346,266
R^2	0.449	0.480	0.456	0.482				
Type × Year trend			✓	✓			✓	✓
Property FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
PC4 × Year FE		✓		✓		✓		✓

Notes: The table reports alternative estimators of the baseline Difference-in-Differences estimates. Columns (1)–(4) use the estimator of Sun and Abraham (2021), while Columns (5)–(8) use the estimator of Borusyak et al. (2024). All specifications are otherwise identical to those in Table 1. Standard errors are clustered at the property level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A4: Alternative Estimators of Wealth Heterogeneity Correction

	Private Rental							
	SA				BJS			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Violation	1.795***	1.389***	1.452***	1.271***	1.684***	1.337***	1.346***	1.217***
	(0.283)	(0.288)	(0.279)	(0.287)	(0.284)	(0.292)	(0.282)	(0.291)
Type × Year trend			1.636***	1.380***				
			(0.050)	(0.068)				
N	364,186	361,115	364,186	361,115	364,179	364,113	364,179	364,113
R^2	0.449	0.479	0.456	0.481				
Type × Year trend			✓	✓			✓	✓
Property FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
PC4 × Year FE		✓		✓		✓		✓

Notes: The table reports alternative estimators of the wealth heterogeneity correction. Columns (1)–(4) use the estimator of Sun and Abraham (2021), while Columns (5)–(8) use the estimator of Borusyak et al. (2024). All specifications are otherwise identical to those in Table 2. Standard errors are clustered at the property level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A5: Alternative Estimators of Household Income Trend Correction

	Private Rental							
	SA				BJS			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Violation	0.725***	0.594***	0.578***	0.497***	0.737***	0.551***	0.611***	0.505***
	(0.131)	(0.137)	(0.129)	(0.136)	(0.137)	(0.132)	(0.137)	(0.134)
Type × Year trend			1.641***	1.403***				
			(0.054)	(0.073)				
N	317,484	314,429	317,484	314,429	316,891	316,891	316,891	316,891
R^2	0.447	0.480	0.454	0.482				
Type × Year trend			✓	✓			✓	✓
Property FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
PC4 × Year FE		✓		✓		✓		✓

Notes: The table reports alternative estimators of the household income trend correction. Columns (1)–(4) use the estimator of Sun and Abraham (2021), while Columns (5)–(8) use the estimator of Borusyak et al. (2024). All specifications are otherwise identical to those in Table 3. Standard errors are clustered at the property level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A6: Robustness Check Using Transactions from 2006–2010

	Private Rental							
	TWFE				CSDID			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Violation	0.946***	0.726***	0.740***	0.604***	0.879***	0.697***	0.788***	0.664***
	(0.146)	(0.147)	(0.145)	(0.147)	(0.150)	(0.149)	(0.149)	(0.149)
Type × Year trend			1.481***	1.285***				
			(0.031)	(0.041)				
N	1,097,536	1,095,771	1,097,536	1,095,771	1,096,100	1,096,220	1,096,100	1,096,220
R^2	0.555	0.567	0.559	0.568				
Type × Year trend			✓	✓			✓	✓
Property FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
PC4 × Year FE		✓		✓		✓		✓

Notes: The table reports robustness checks using transactions from 2006–2010 to construct the baseline sample. Columns (1)–(4) report estimates from two-way fixed-effects (TWFE) regressions, while Columns (5)–(8) use the estimator of Callaway and Sant'Anna (2021). All specifications include property and year fixed effects. Columns (2), (4), (6), and (8) include PC4-by-year fixed effects, while Columns (3), (4), (7), and (8) additionally control for type-specific linear time trends. Standard errors are clustered at the property level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A7: Buy-to-Let and Keep-to-Let

	Private Rental with Transaction				Private Rental without Transaction			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Violation	0.557***	0.362**	0.335**	0.243	0.763***	0.637***	0.628***	0.589***
	(0.149)	(0.155)	(0.149)	(0.155)	(0.204)	(0.211)	(0.203)	(0.211)
Type × Year trend			1.029***	0.989***			0.624***	0.397***
			(0.038)	(0.052)			(0.038)	(0.051)
N	346,470	343,336	346,470	343,336	346,470	343,336	346,470	343,336
R^2	0.375	0.410	0.383	0.413	0.457	0.483	0.458	0.483
Type × Year trend			✓	✓			✓	✓
Property FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
PC4 × Year FE		✓		✓		✓		✓

Notes: The table examines the effect of borrowing constraints on two types of conversions from owner-occupied housing to private rental housing: buy-to-let and keep-to-let. The dependent variable in Columns (1)–(4) is Private Rental with Transaction, while that in Columns (5)–(8) is Private Rental without Transaction. All specifications include property and year fixed effects. Columns (2), (4), (6), and (8) include PC4-by-year fixed effects, while Columns (3), (4), (7), and (8) additionally control for type-specific linear time trends. Standard errors are clustered at the property level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A8: PC6-Level Violation Probability and Private Rental

	Private Rental			
	(1)	(2)	(3)	(4)
PC6-Year Violation Rate	0.667***	0.593***	0.360***	0.300***
	(0.121)	(0.117)	(0.109)	(0.107)
Type × Year trend			1.562***	1.540***
			(0.0214)	(0.0251)
N	3,362,187	3,362,180	3,362,187	3,362,180
R^2	0.500	0.503	0.505	0.507
Type × Year trend			✓	✓
Property FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
PC4 × Year FE		✓		✓

Notes: The table examines the relationship between borrowing constraints at the PC6-year level and the probability that a property is privately rented. PC6-Year Violation Rate is the average violation probability at the PC6-year level. The sample includes all property-year observations from 2011 to 2017 in PC6 areas with transactions in 2009 and 2010. All specifications include property and year fixed effects. Columns (2) and (4) include PC4-by-year fixed effects, while Columns (3) and (4) additionally control for type-specific linear time trends. Standard errors are clustered at the property level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

C.2 Additional figures

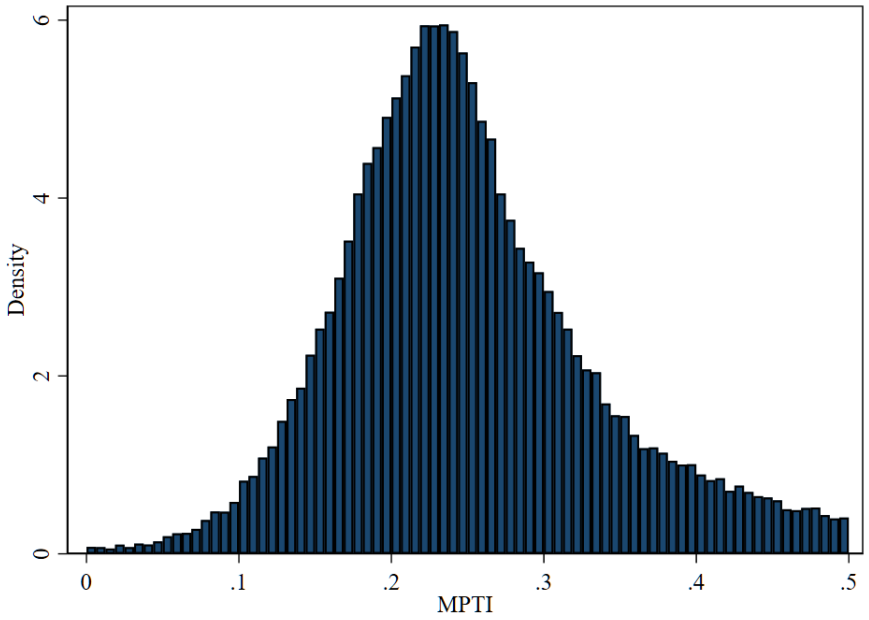


Figure A3: Distribution of Mortgage Payment-to-Income Ratios

Notes: This figure presents the distribution of mortgage payment-to-income (MPTI) ratios among first-time buyers of owner-occupied properties in 2009 and 2010, corresponding to the sample summarized in Table A1. Observations with an MPTI ratio equal to zero or greater than 0.5 are excluded.

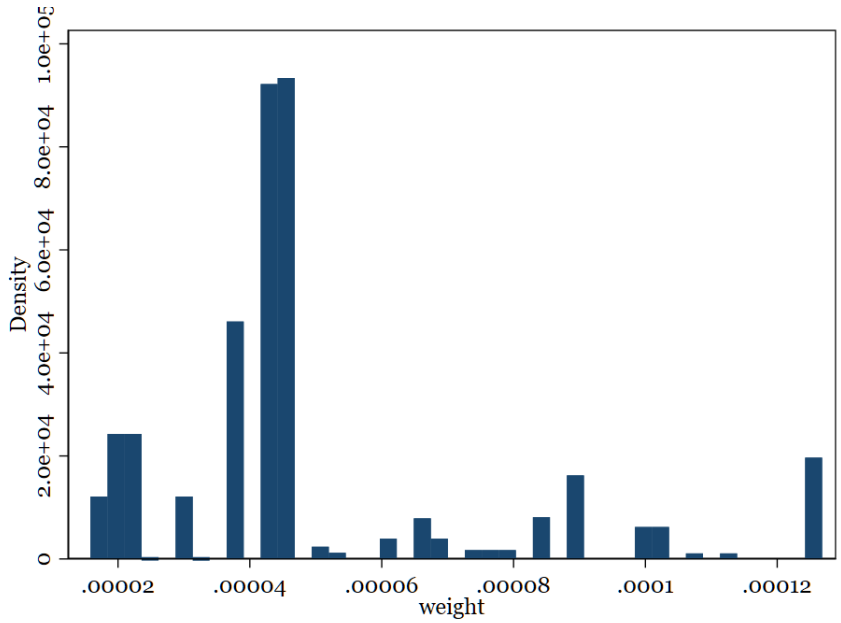
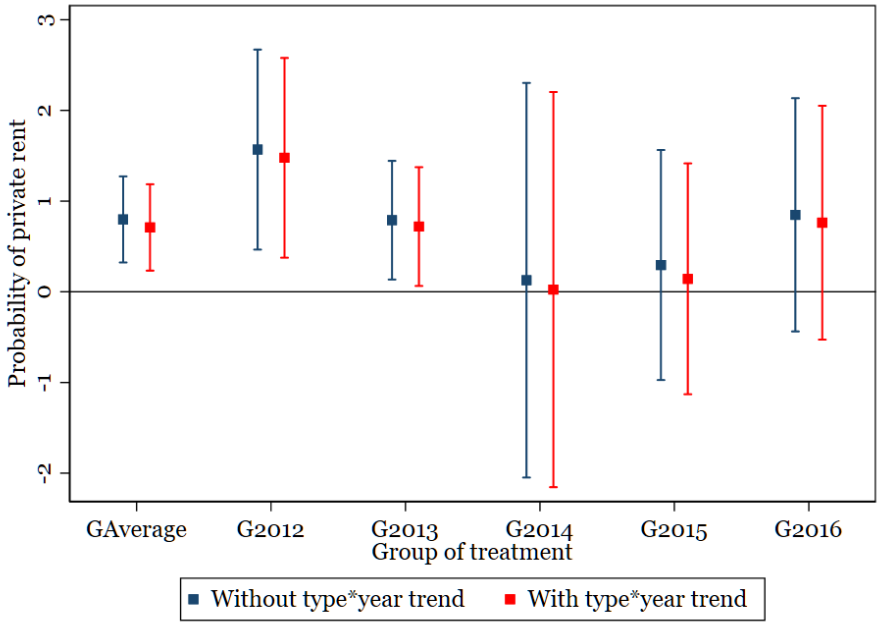
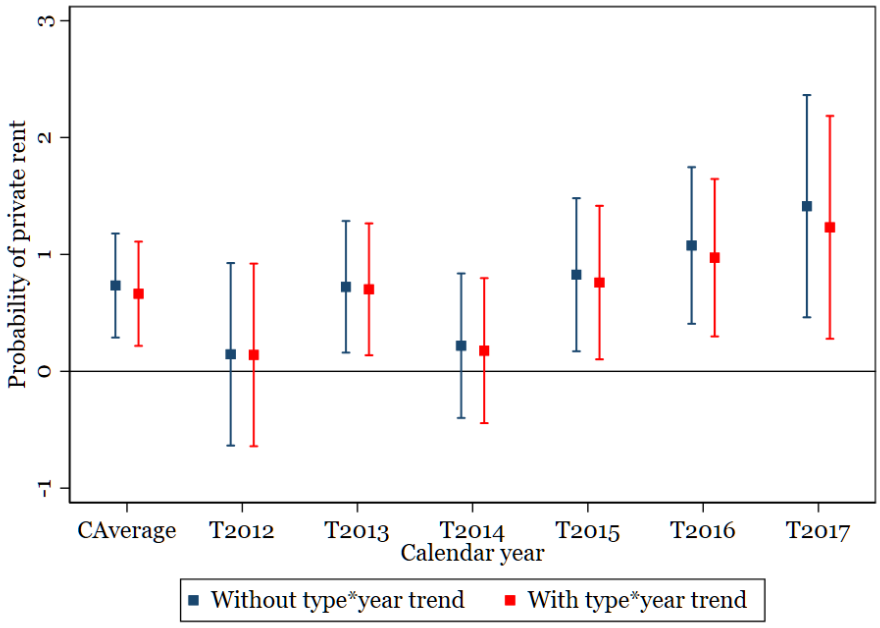


Figure A4: Distribution of Treatment Weights in the TWFE Estimator

Notes: This figure presents the distribution of the weights assigned to treatment effects in the two-way fixed effects estimator. The vertical line indicates the average weight. The weights are calculated using the *twowayfeweights* command in Stata 16.0.



(a) Treatment effects by treatment cohort



(b) Treatment effects by calendar year

Figure A5: Treatment Effect Heterogeneity across Cohorts and Calendar Years

Notes: This figure presents the decomposition of treatment effects following Callaway and Sant'Anna (2021). Panel (a) reports treatment effects by treatment cohort, while Panel (b) reports treatment effects by calendar year. The cumulative treatment effect by the end of 2017 ranges from 1.232 to 1.413.

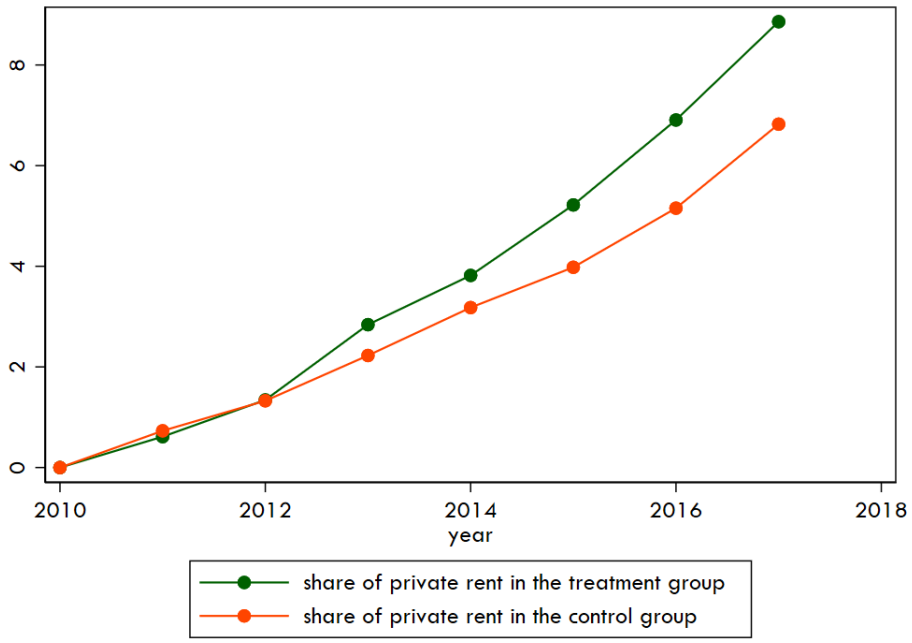
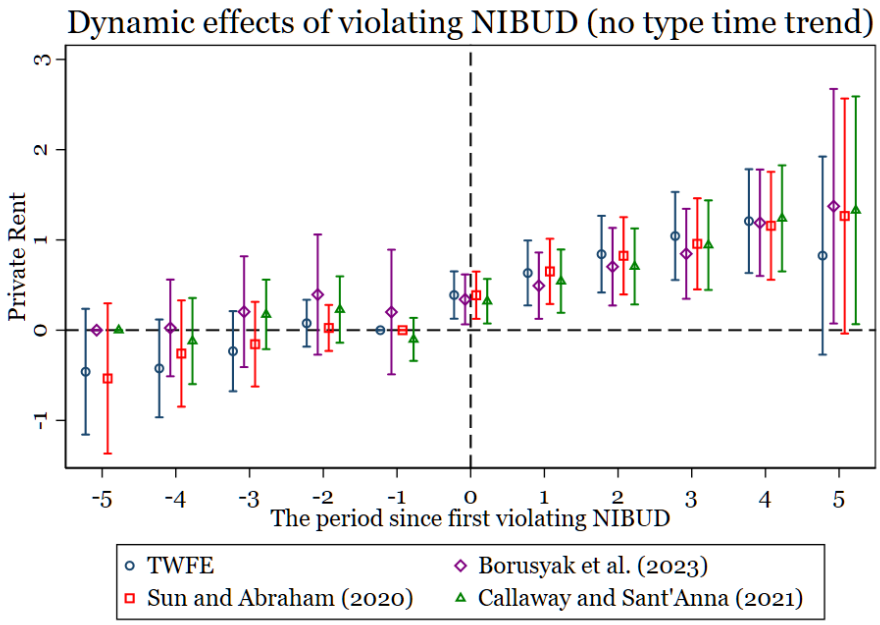
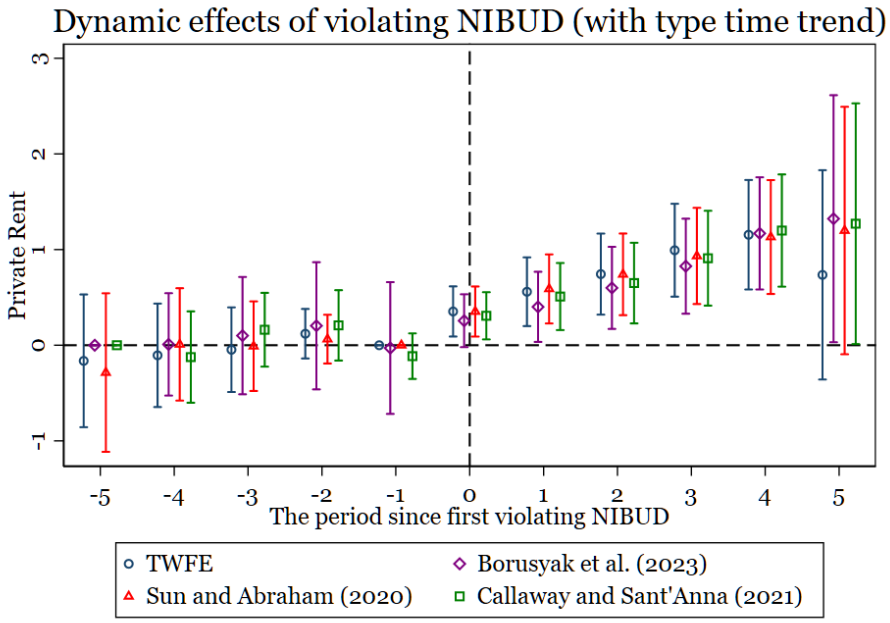


Figure A6: Evolution of Private Rental Shares in the Treatment and Control Groups

Notes: This figure shows the evolution of the share of properties in the private rental sector for the treatment and control groups. The treatment group consists of properties that violate the maximum MPTI constraint between 2011 and 2017, while the remaining properties form the control group. All properties are owner-occupied at the end of 2010. The private rental share subsequently increases more rapidly in the treatment group, reaching 8.86% by the end of 2017, compared with 6.82% in the control group.



(a) Without type-specific linear time trends



(b) With type-specific linear time trends

Figure A7: Dynamic Effects in the Extended Sample

Notes: This figure presents dynamic treatment effect estimates with 95% confidence intervals using the extended sample, based on equation (18). Panel (a) reports estimates without type-specific linear time trends, while Panel (b) includes them, corresponding to Columns (2) and (4) of Table A6, respectively. Standard errors are clustered at the PC4 $\times$ year level.

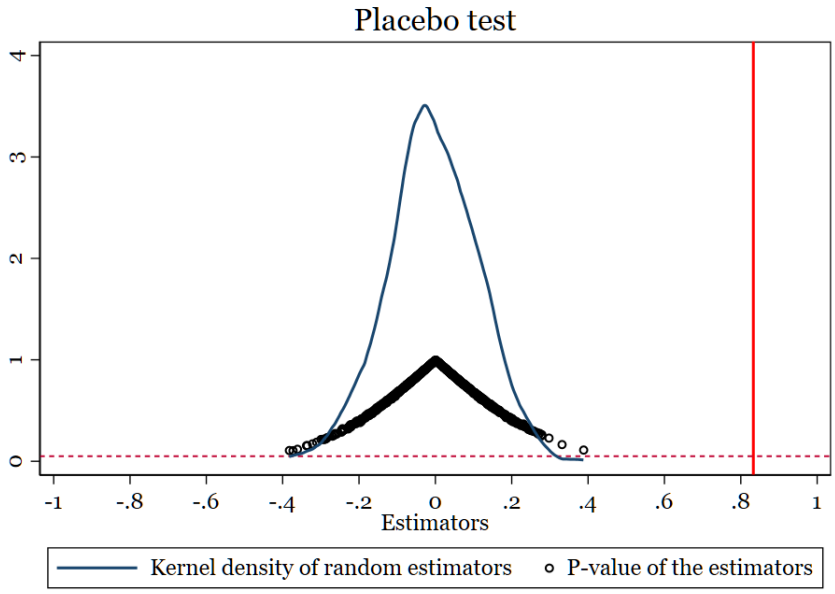


Figure A8: Randomization Placebo Test

Notes: This figure presents a randomization placebo test based on 1,000 fictitious treatment assignments. In each iteration, a placebo treatment group is randomly drawn from the original sample and the corresponding treatment effect and p-value are estimated. The figure shows the resulting distribution of placebo estimates, while the vertical red line indicates the actual estimated treatment effect of 0.833 in column (4) of Table 1.

D The 2011 harmonization in the Code of Conduct

As an example of the Code of Conduct that was applicable before 2011, we refer to the Code that became effective January 1, 2007, the *Gedragscode Hypothecaire Financieringen (1 januari 2007)* (Nederlandse Vereniging van Banken, 2007). In this code, sub-section 6.6 referred to the exceptions to the NIBUD rule, and in translation reads:

"6. When providing a mortgage loan the mortgage lender may, in special cases, exceed the standards referred to above under sub-sections 2, 3, and 4, provided that the lender has informed the consumer of this exceedance in a timely manner, and the consumer has provided a written statement to the mortgage lender acknowledging that the lender has pointed out the exceedance of the standards and the associated risks, and that the consumer understands and accepts these risks. The mortgage lender must document the exceedance and the underlying justification in the consumer's financing file."

For completeness, the referenced sub-sections 2, 3 and 4 read in translation:

"2. When determining the borrowing capacity of a borrower applying for a mortgage loan, the mortgage lender will take account of current fixed and long-term income, as well as any income available to the borrower in the future from disposable capital, provided this is reasonably foreseeable. "Income available to the borrower in the future from disposable capital" means income which the borrower may spend without affecting any of the capital from which the income is derived. When determining this income, the mortgage lender will, at the most, base its calculation on a percentage to be set by the Mortgage Lenders' Contact Body, that is now set at 3% of the value of the capital. If the interest rate percentage applicable to a mortgage loan is less than the above percentage the mortgage lender will, at the most, base its calculation on the interest rate percentage applicable to the mortgage loan when defining the income at the borrower's disposal. If the mortgage loan is provided to more than one borrower, the mortgage lender may take account of their joint incomes when determining their borrowing capacity.

3. The mortgage lender will determine the maximum amount of the gross expenses linked to a mortgage loan based upon the current housing costs set by the National Institute for Family Finance Information (NIBUD) in Utrecht. If the mortgage loan is provided to more than one borrower, the housing cost percentage is based on the borrower with the highest income.

4. The mortgage lender will calculate the borrower's borrowing capacity for the purposes of a mortgage loan with a fixed interest rate period lasting for less than ten years based on the percentage set by the Mortgage Lenders' Contact Body based on the market interest rate paid on 10-year Dutch state bonds, plus a surcharge determined by the Mortgage Lenders' Contact Body. When calculating a borrower's borrowing capacity the mortgage lender may also use a higher

notional interest rate.

The mortgage lender will calculate the borrower's borrowing capacity for the purposes of a mortgage loan with a fixed interest rate period lasting for ten years or longer based on the interest rate percentage that the mortgage lender in fact charges in the course of that fixed interest rate period. Irrespective of the method of repayment or the fixed interest rate period of the mortgage loan the borrowing limit will, at the least, be based upon the costs of a 30-year annuity loan."

With the Code of Conduct that became effective on August 1, 2011 (Nederlandse Vereniging van Banken, 2011), the sub-section 6.6 on exceptions to the NIBUD rule was changed to:

"6. When providing a mortgage loan the mortgage lender may, in the following cases, depart from the provisions set out in sub-sections 2, 3, 4 and 5:

a. when remortgaging a mortgage loan on condition that the borrower continues to live in the residence and the new mortgage loan does not exceed the mortgage loan to be repaid, plus the notarial and valuation fees, possible settlement charges and/or payments paid by the borrower to a broker in respect of this mortgage loan and such charge as may have been incurred as a result of an early redemption of the previous mortgage loan;

b. in the case of energy-saving facilities in the residence and when acquiring a residence bearing what is termed an A label, an additional mortgage loan may be advanced up to an amount to be set by the Mortgage Lenders' Contact Body in consultation with NIBUD;

c. in a 'bridging situation' the borrower's obligations under an existing mortgage loan covering a residence that has been, or will be, sold and under a bridging loan may be disregarded provided it is likely that the borrower will be able to meet the obligations under the mortgage loans over a realistic period."

The referenced sub-sections 2, 3, 4 and 5 contain the exact same paragraphs as in 2007 (although in a slightly different order), except for the addition of the following paragraphs in sub-section 3:

"When determining borrowing capacity, account may also be taken of:

...

b. an increase in income based on an unconditional written undertaking by an employer to a borrower if this increase in income is to be effected within 6 months of the date of that undertaking. In this case, the income may be supplemented by the full amount of the future increase.

c. the average income over the last three calendar years prior to the year in which the mortgage loan is offered or advanced where the borrower is an entrepreneur

who enjoys no fixed and long-term income.
If the borrower has enjoyed a fixed and long-term income for less than three calendar years the mortgage lender may, in addition to the available calendar years, base its calculation on a forecast of the borrower's future income drawn up by a suitably qualified professional adviser."